Tracing Semantic Change in ADHD and Autism Discourse at Web Scale

Jakob Lütkemeier

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Ethics and Data-Handling Statement

This study involved no human participants, no interaction or intervention, and no primary data collection; it is a secondary analysis of publicly accessible web pages archived and distributed by Common Crawl. The corpus was obtained in accordance with Common Crawl's terms of use, and the archive itself honours publishers' `robots.txt` exclusions; no attempt was made to access or reconstruct content excluded from the crawl. Although the analysed texts include first-person accounts of mental health experiences, all analyses were conducted at the level of aggregate annual corpus statistics. No individuals were identified or profiled, no author- or user-level inferences were drawn, and no personal identifiers were extracted or analysed beyond their incidental presence in archived page text. Any example passages shown are brief and non-identifying. Processing followed the data-protection principles of purpose limitation and data minimisation: collection artefacts were stored on access-controlled cloud infrastructure (AWS S3/EC2) and used solely for this research, and the released reproducibility materials comprise code, configuration, and derived aggregate outputs, with the underlying web documents referenced by their Common Crawl identifiers rather than redistributed.

Statement on the Use of Artificial Intelligence

Generative AI tools were used in three capacities in this project. First, as coding assistants: OpenAI's ChatGPT Codex was used to help implement and debug parts of the software developed for this study, most extensively the Common Crawl text-extraction pipeline; all code was specified, reviewed, tested, and validated by the author, who takes full responsibility for its correctness. Second, as research instruments: the frame-annotation workflow described in the Methods chapter uses large language models by design (OpenAI Codex GPT-5.5 as annotator and Gemini 3 Flash Preview as critic), with human adjudication authoritative throughout; this methodological use is documented in full in Section 4.2. Third, as a language aid: bar minor polishing of prose, grammar and wording, all text was written by the author. AI tools were not used to search, select, or review the literature, and all interpretations and conclusions are the author's own. The author accepts full responsibility for the entire content of this dissertation.

Abstract

Mental health language has permeated everyday discourse, prompting concern that diagnostic terms are being trivialised and drained of meaning. Concept creep theory predicts that harm-related concepts undergo lexical semantic change (LSC), broadening horizontally and becoming vertically milder, yet existing evidence rests almost entirely on curated corpora and rarely separates change in meaning from change in discourse composition. This dissertation shifts the evidentiary base to general web discourse, asking whether ADHD and autism—two diagnostic concepts at the centre of contemporary debates about therapy-speak and prevalence inflation—have undergone the predicted semantic loosening.

A reproducible pipeline extracted quality-filtered English documents mentioning ADHD, autism, and three baseline emotion terms (*frustration*, *sadness*, *loneliness*) from thirteen annual Common Crawl snapshots spanning 2014–2026, yielding 96,864 target contexts. A validated hierarchical classifier, developed through LLM-assisted annotation under human adjudication, separated clinical/disorder from lived-experience framing. Adapting the SIBling framework, annual trajectories were estimated for salience, sentiment (NRC–VAD valence), intensity (NRC–VAD arousal), contextual breadth (XL-LEXEME target-aware embeddings), and thematic neighbour similarity, overall and within frames, with robustness checks against the framework's original operationalisations.

Neither hypothesised form of concept creep was supported. Intensity did not decline in any target stratum, breadth contracted rather than broadened under the target-aware operationalisation—most markedly for ADHD—and the thematic core of both discourses remained diagnostic and child-centred throughout. The clearest change was instead compositional: discourse shifted robustly from clinical toward lived-experience framing, with the ADHD lived-experience share among clear frames rising from a fitted 17.5% to 30.3%, accompanied by a recent positive valence shift in clinical contexts and declining autism salience in the sampled crawl. Robustness checks showed that the vertical-creep conclusion depends on the affective lexicon and that breadth trends depend on encoder choice. The findings indicate semantic consolidation rather than concept creep: these institutionally codified categories may have expanded to encompass more people (through looser DSM criteria), while the meaning of their labels in use has largely held its shape. Discourse composition and measurement validation emerge as central methodological concerns for LSC research.

Acknowledgements

I would like to thank my supervisor, Dr Tom Paskhalis, for guidance throughout this dissertation, and the MSc in Applied Social Data Science teaching team for methodological training and support.

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1 Introduction

Popular and scholarly accounts alike describe an unfolding mental health crisis across many countries. Evidence of escalating mental health complaints comes from surveys, epidemiological studies, and rising prescriptions of psychiatric medications (Haidt, 2025). Particularly substantial increases in prevalence have been reported for the neurodevelopmental disorders autism (Zeidan et al., 2022) and attention-deficit hyperactivity disorder (ADHD) (Cui et al., 2026; McKechnie et al., 2023). In Australia, for example, the number of people with autism rose by 42% between 2018 and 2022, including a 95% increase among women and girls (Australian Bureau of Statistics, 2023). In the United Kingdom, ADHD diagnoses increased twofold among boys, threefold among girls, and more than tenfold among adults between 2000 and 2018 (McKechnie et al., 2023).

Concurrently, mental health awareness campaigns have proliferated (Bolinski et al., 2020; Sampogna et al., 2017), and mental-health-related discourse has become an increasingly ordinary feature of everyday life (Foulkes and Andrews, 2023; Medaris, 2024). Here too, ADHD and autism are particularly stark examples. By June 2026, ADHD and autism had been hashtagged in more than 5.4 million and 4 million TikTok videos, respectively [1], and an analysis of a LexisNexis dataset identified 25,080 media articles on ADHD between January and May 2024, compared with 5,775 in the equivalent period of 2014 (Martin et al., 2025).

On the one hand, this popularisation of mental-health-related discourse has genuine benefits. It may give people language for subtle emotional states (Medaris, 2024), improve mental health literacy (Schomerus et al., 2012), and thereby reduce stigma around mental illness (Sampogna et al., 2017; Fleary et al., 2022). It may also support better recognition and more accurate reporting of mental health problems (Foulkes and Andrews, 2023), erode barriers to help-seeking that contribute to the ongoing under-treatment of some conditions (Haslam and Tse, 2026), and facilitate online identity and community formation, particularly around ADHD and autism (Ginapp et al.,

[1] Hashtag video counts as displayed on TikTok's search page, retrieved by the author on June 29, 2026

2023).

On the other hand, the concurrence of rising prevalence and proliferating discourse has prompted scholars to interrogate the relationship between the two (Haslam, 2026; Foulkes and Andrews, 2023). Foulkes and Andrews (2023) term this account the *prevalence inflation hypothesis*, which posits a cyclical, mutually reinforcing dynamic: on one side of the cycle, rising prevalence may promote the permeation of mental health terminology into mainstream discourse; on the other, the increased availability of such discourse may itself inflate prevalence by encouraging overdiagnosis, overinterpretation, or pathologisation, whereby milder or more transient forms of distress come to be understood and reported as clinical mental health problems, often in conjunction with self-diagnosis (Foulkes and Andrews, 2023; Beeker et al., 2021; Haslam, 2026).

The mechanism is not merely linguistic. Once individuals interpret and label their psychological experiences as symptoms of a mental health condition, self-concept and behaviour may change in turn (Foulkes and Andrews, 2023; Alper et al., 2025). In stronger cases, such labelling may become self-fulfilling: ordinary distress, reframed as symptomatic of disorder, may give rise to patterns of attention, avoidance, identification, or help-seeking that intensify the very symptoms being labelled (Foulkes and Andrews, 2023; Haslam, 2026). This process may be further amplified where mental health problems are glamorised or romanticised. Under these conditions, diagnostic and quasi-diagnostic labels may acquire social value, making self-labelling attractive rather than merely explanatory (Ndour and Foulkes, 2025).

These concerns are particularly acute for ADHD and autism, for which clinicians have reported increases in self-diagnosed presentations (Hartnett and Cummings, 2024; Weigle and Shafi, 2024), and they may be especially pronounced among adolescents, given their heightened susceptibility to rumination, peer influence, identity formation, social reward, and media messaging (Foulkes and Andrews, 2023; Ndour and Foulkes, 2025).

A further corollary of the diffusion of mental health language into popular culture is the phenomenon of *therapy-speak*, “the imprecise and superficial integration of psychotherapy language into everyday communication” (Isern-Mas and Almagro, 2025). Its consequences are twofold. First, therapy-speak may spread inaccurate psychological information. An analysis of the 100 most popular TikTok videos about ADHD found that more than half (55%) of the characteristics attributed to ADHD by video creators did not align with the diagnostic criteria of the fifth revision of the *Diagnostic and Statistical Manual of Mental Disorders* (DSM-5) (de Vries, Batstra, and van Assen, 2025); similarly, of the 133 most-viewed TikTok videos tagged #autism, only 27% were rated as accurate, 41% as inaccurate, and 32% as containing potentially misleading overgeneralisations (Aragon-Guevara et al., 2025). Second, therapy-speak may erode the meaning and relevance of mental-health-related terms, a process that can also be understood as *trivialisation*. This can contribute to hermeneutical injustice, depriving people who genuinely live with a condition of the words they need to describe their experiences (Isern-Mas and Almagro, 2025) and reducing their symptoms to mere personality traits, thereby denying them a fully recognised psychiatric identity (Spencer and Carel, 2021).

This latter concern—the erosion of meaning in psychotherapy terms—is the focus of the present study, which investigates lexical semantic change (LSC) in the terms ADHD and autism in general web discourse. The literature review that follows first synthesises empirical evidence on LSC in mental-health-related terms over recent decades, before reviewing the computational approaches used to study these processes and identifying the gaps addressed by this project.

2 Related Work

Lexical semantic change (LSC) refers to shifts in a word's meaning while its grammatical function remains stable, and it constitutes a common form of language change (Campbell, 2013). *Mouse*, for example, initially a word for a small rodent, broadened in usage to denote a computer input device as well.

Concept creep theory posits that harm-related concepts are particularly prone to LSC, specifically to gradual semantic broadening; accordingly, many harm-related terms—including *addiction*, *bullying*, *harassment*, *prejudice*, and *trauma*—are reported to have expanded in meaning since at least the 1970s (Haslam, 2016; Baes et al., 2023). The theory distinguishes two forms of LSC that can occur concurrently: vertical creep, the loosening of definitions to include milder instances (declining semantic severity or intensity), and horizontal creep, the extension of definitions to encompass qualitatively new phenomena (semantic broadening). Concepts of mental illness qualify as harm-related because distress and dysfunction are fundamental to their definition (Wakefield, 1992), and a series of studies has therefore examined concept creep across mental-health-related concepts.

Vylomova and Haslam (2021) found that *trauma* and *addiction* (together with three concepts not specific to mental health: *bullying*, *prejudice*, and *harassment*) exhibited both vertical and horizontal creep in a corpus of approximately 800,000 psychology article abstracts from 875 journals dating back to the 1960s. Similar, though weaker, patterns emerged in a general-language corpus combining the Corpus of Contemporary American English (COCA) and the Corpus of Historical American English (COHA) from the 1970s to the 2010s. Subsequent studies using the same corpora and time periods report comparable findings for other mental-health-related concepts: Baes et al. (2023) found evidence of vertical creep in *trauma* in the psychology corpus; Baes, Haslam, and Vylomova (2023) reported vertical creep in *addiction*, *anger*, *stress*, and *worry* in the psychology corpus, and in *addiction*, *grief*, *stress*, and *worry* in the general corpus; and Baes, Haslam, and Vylomova (2024) further found that the broader concepts *mental health* and *mental illness* both expanded horizontally in the psychology corpus.

Not all findings fit this pattern. One study reported that the semantic intensity of *anxiety* and *depression* increased in both Vylomova and Haslam's psychology corpus and their general-language corpus, contrary to concept-creep expectations (Xiao et al., 2023). This unexpected result may reflect insufficiently specific measurement, in particular the absence of discourse-context and construct distinctions: *depression* may denote psychiatric depression but also meteorological or economic phenomena, and *anxiety* may refer to a nosological category, a disorder construct, or an underlying human experience. In a replication study, Pisl, Bucur, and Podina (2025) showed that this apparent increase in semantic intensity could be attributed to changing discourse composition—shifts in the balance between clinical or nosological contexts and lived-experience contexts—rather than to intrinsic semantic change alone. In their analysis of lead paragraphs from *New York Times* articles published between 1970 and 2023, the time effect for *depression* became nonsignificant after controlling for mental-health context.

Another study examined a range of terms commonly associated with therapy-speak, including *toxic*, *bipolar*, *psychopath*, *narcissistic*, and *triggered*, and reported mixed results (Iacob and Uban, 2026). In an extension of Vylomova and Haslam's psychology corpus, most terms exhibited horizontal creep; by contrast, in two Reddit corpora comprising comments from psychology-oriented and general subreddits between 2010 and 2025, most terms narrowed in breadth. Overall, the authors found that long-established psychological terms such as *OCD*, *bipolar*, and *trauma* displayed little semantic change, whereas more vernacular terms such as *gaslighting* and *imposter* shifted substantially from year to year.

On balance, many mental-health-related concepts appear to have undergone concept creep, becoming milder and broader over recent decades. Haslam and colleagues theorise several drivers of this process (Haslam et al., 2020; Haslam, 2026). One is the influence of "opprobrium entrepreneurs", who seek to cast previously accepted conditions in a more problematic light: broadening a harm concept allows the disapproval associated with its original meaning to extend to less severe cases. A second is "prevalence-induced conceptual change", whereby the standards for recognising harm tend to loosen as instances of harm become less common. The third is a broader cultural increase in concern with harm: as harm concepts expand, a wider range of experiences is treated as morally significant and worthy of care, and a greater number of actions come to be seen as harmful (Haslam et al., 2020; Haslam, 2026).

Whether this broadening of mental-health constructs reflects changes in official diagnostic criteria is unclear. A meta-analysis showed that no revision of the *Diagnostic and Statistical Manual of Mental Disorders* (DSM) from the third edition onward was reliably more inflationary or deflationary overall. Specific disorders, however, changed significantly: most notably, ADHD inflated by 18%, 33%, and then 17% across the three revisions following DSM-III, while autism inflated by 50% from DSM-III to DSM-III-R—albeit on the basis of a single study—but deflated by 15% from DSM-IV to DSM-5 (Fabiano and Haslam, 2020).

2.1 Computational Approaches to Studying LSC

Semantic change has long been studied across linguistics and the social sciences, yet lexical semantic change remains difficult to characterise: shifts in word meaning are often gradual and less visible than other forms of linguistic change, such as those produced by spelling or grammar reforms (de Sá, Silveira, and Pruski, 2026). Earlier research largely depended on manual methods, with linguists drawing on historical texts, dictionaries, and corpora to reconstruct how word meanings changed over time. More recently, computational linguistics and natural language processing (NLP) have made it possible to study semantic change at much larger scales (Jurafsky and Martin, 2026). Within NLP, meaning is typically understood in distributional terms—a word's meaning is inferred from the contexts in which it appears—and these contextual patterns can be represented through several computational approaches, including frequency-based measures, topic models, semantic graphs, and embedding-based methods (de Sá, Silveira, and Pruski, 2026).

Although considerable progress has been made in identifying LSC with these techniques (see Tahmasebi, Borin, and Jatowt, 2019; Kutuzov et al., 2018; Tang, 2018), less attention has been paid to characterising the nature of the changes identified. To address this gap, Baes, Haslam, and Vylomova (2024) proposed the SIBling framework—Sentiment, Intensity, and Breadth—as a unified, multidimensional approach to characterising LSC. The framework situates concept creep within a broader account of LSC and links this conceptual model to computational methods, including frequency-based and embedding-based techniques. It distinguishes three dimensions along which terms may change over time: vertical drift, horizontal drift, and sentiment. Vertical drift refers to changes in intensity: a term may strengthen, as in *hilarious* shifting from cheerful or amusing to extremely funny, or weaken, as in *trauma* shifting from brain injuries to milder events such as business loss; this dimension maps onto vertical concept creep. Horizontal drift refers to changes in breadth: a term may narrow, as in *doctor* shifting from scholar or teacher to primarily denoting a medical professional, or broaden, as in *cloud* shifting from a meteorological term to internet-based data storage; this dimension maps onto horizontal concept creep. Sentiment refers to changes in connotation: a term may acquire a more positive connotation, as in *geek* shifting from a derogatory term for odd people to someone passionate about a field, or a more negative one, as in *retarded* shifting from a neutral term for intellectual disability to a highly pejorative slur; this dimension corresponds roughly to destigmatisation and stigmatisation, which are not directly captured by concept creep theory. According to SIBling, these dimensions can be complemented by examining shifts in target-word frequency, or salience, and in the thematic content of target-word collocates. Despite its conceptual value, however, the framework's operationalisations should be treated as first implementations rather than settled measures: they have not yet been extensively validated externally, and their measurement choices remain open to refinement (Baes, Haslam, and Vylomova, 2024).

2.2 The Present Study

Three gaps in the existing literature motivate the present study.

The first concerns the evidentiary basis of prior work. Research on LSC in mental-health-related concepts has relied heavily on a small set of curated corpora, particularly COCA/COHA and Vylomova and Haslam's (2021) psychology abstracts datasets (e.g., Baes, Haslam, and Vylomova, 2023, 2024; Xiao et al., 2023; Iacob and Uban, 2026). This matters because the broader aim of this literature is to infer cultural dynamics from lexical change, yet curated corpora may partly reflect shifts in editorial policy, audience composition, disciplinary conventions, or ideological stance rather than broader changes in public discourse (Pisl, Bucur, and Podina, 2025). To date, LSC in mental-health-related concepts has not been examined systematically in general web discourse.

The second concerns the target concepts themselves. Direct evidence on LSC in ADHD and autism remains limited. One study found that the two concepts have converged semantically on Reddit: from 2019 onward, their contextual similarity increased, and the terms became more similar to each other than to comparison conditions (Kang, Haslam, and Conway, 2025). This finding, however, captures convergence between ADHD and autism on a single platform; it does not characterise how either concept's semantic profile has evolved over time in general web discourse.

The third concerns discourse composition. Existing work indicates that apparent trends in LSC may partly reflect shifts in the contexts in which terms are used—such as clinical categories, service labels, identity or lived-experience categories, and objects of public debate—rather than intrinsic semantic change (Pisl, Bucur, and Podina, 2025; Xiao et al., 2023). Treating all mentions as a single semantic population therefore risks conflating genuine semantic change with change in discourse composition.

Taken together, these gaps point to three corresponding requirements: a broader evidentiary base that captures general web discourse, direct diachronic analysis of ADHD and autism, and an analytic approach that accounts for variation in discourse framing. The present study was designed to meet these requirements and addresses four research questions:

RQ1: How has the salience of ADHD and autism in general web discourse changed from 2014 to 2026, against the backdrop of non-clinical negative-emotion baseline terms?

RQ2: How has the balance between clinical and lived-experience framing of ADHD and autism changed over this period?

RQ3: How have the intensity, breadth, and sentiment of ADHD and autism contexts changed—overall and within clinical versus lived-experience frames—against the backdrop of the baseline terms?

RQ4: How has the thematic content of ADHD and autism discourse evolved over time?

Consistent with concept creep theory, it is hypothesised that ADHD and autism will increasingly appear in less emotionally intense contexts (vertical concept creep) and across a broader qualitative range of lexical contexts (horizontal concept creep). This hypothesis corresponds to the intensity and breadth components of RQ3; the remaining measures—sentiment (RQ3), frame composition (RQ2), and thematic content (RQ4)—are treated as exploratory, given the limited prior research available to motivate directional predictions.

Methodologically, the study adapts Baes et al.'s (2024) SIBling framework for LSC analysis, refining its operationalisation through target-aware embeddings for breadth and an expanded affective lexicon for intensity and sentiment. To broaden the evidentiary base and support direct diachronic semantic analysis of ADHD and autism, it constructs a diachronic corpus of general web discourse using Common Crawl, a large-scale, openly accessible repository of web crawl data. The corpus spans 2014–2026, from the earliest period with sufficiently consistent and usable data to the most recent available year. An efficient and reproducible pipeline extracts quality-filtered web documents containing ADHD, autism, and baseline emotion terms, supporting both frequency estimation and downstream semantic analysis while ensuring comparability across targets and baselines. Finally, to account for variation in discourse framing, the study incorporates a supervised frame-classification step that distinguishes clinical or disorder-construct discourse from lived-experience discourse prior to LSC analysis.

In sum, the present study contributes to the literature by extending concept creep and therapy-speak research to ADHD and autism; by shifting the evidentiary base from curated corpora toward general web discourse; by introducing a frame-aware approach that separates clinical from lived-experience uses of mental health terms; by implementing a reproducible Common Crawl pipeline for LSC research; and by assessing whether selected operational refinements can strengthen the application of the SIBling framework.

3 Data and Materials

3.1 Common Crawl

Common Crawl is the largest freely available public archive of web crawl data, totalling more than 10 petabytes across crawls published approximately monthly, each typically containing more than two billion web pages (Baack, 2024; Common Crawl Team, 2024). Widely used as a web-scale source of language data for corpus construction, NLP research, and large language model pretraining (Baack, 2024; Wenzek et al., 2019), it has been cited in over 12,000 research papers (Common Crawl Team, 2024). For the present study, its central value lies in breadth: rather than drawing on a single platform, newspaper archive, or curated corpus, Common Crawl provides repeated cross-sections of general web discourse. The archive stores raw HTML and, importantly, collects only a sample of pages from each domain it visits—meaning, for instance, that only a subset of Wikipedia articles will appear, not the full site. Domain selection and page depth are governed by *harmonic centrality*, a graph-based scoring method adopted in 2017 whereby a domain's importance is determined by the volume of direct and indirect inbound links from other domains, with direct links weighted most heavily; higher-scoring domains are both more likely to be crawled and to have more of their pages fetched (Baack, 2024). Despite its enormous size, Common Crawl is neither a complete copy of the web nor a representative sample thereof. A growing number of rights holders—including major outlets such as the New York Times—now block Common Crawl via the `robots.txt` standard, largely in response to AI training data concerns, and large social media platforms such as Facebook have done so for considerably longer (Baack, 2024). The data used in this study should therefore not be treated as a representative survey of the web or of public opinion; they reflect what Common Crawl crawled, retained, and made available within these structural constraints.

Common Crawl Collection Pipeline

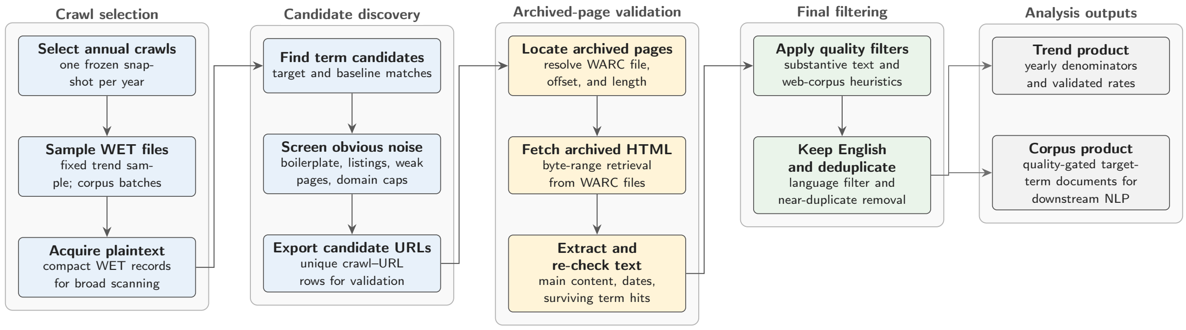

Figure 3.1: Common Crawl collection pipeline used to construct the trend and corpus materials.

To collect the data for this study, a Common Crawl collection pipeline was developed to extract quality-gated general discourse for the analysis of lexical semantic change (LSC) in specific target terms, here ADHD and autism, against matched baseline terms. The pipeline workflow is illustrated in Figure 3.1. Target and baseline terms are processed together so that yearly denominators, sampling logic, and quality filters remain comparable across term groups. The pipeline uses Common Crawl's two main file formats sequentially. WET files provide compact extracted plaintext and are therefore used for large-scale term scanning and yearly prevalence denominators, but they lack the HTML structure and metadata needed for stronger boilerplate filtering. Candidate documents are therefore resolved to their corresponding WARC records, which preserve the full HTML code. Although WARC processing is slower, it enables main-text extraction, boilerplate removal, and metadata recovery. This WET-first, WARC-second design keeps the pipeline economical by reserving expensive WARC processing for candidate documents only.[1]

The design consists of two linked tracks. The *trend track* uses fixed-effort annual samples to estimate how frequently target and baseline terms appear over time. The *corpus track* builds a larger, quality-gated document corpus for downstream NLP analysis. As an additional safeguard against overrepresentation by large websites beyond Common Crawl's restrictive sampling approach, domain caps are applied at 50 WET-validated hit rows per registered domain per Common Crawl crawl. Intermediate summaries and manifests are retained so that each year, crawl, track, and batch remains auditable. The pipeline is designed to run end-to-end on AWS EC2, using S3 as the durable storage and transfer layer for intermediate and final collection artefacts. Yearly crawl selection is deterministic: one Common Crawl snapshot is selected per

[1]For a detailed record of the design, configuration, and methodological decisions behind the Common Crawl collection pipeline refer to `https://github.com/jako6f/msc-nlp-therapy-speak/blob/main/reports/commoncrawl_corpus_design_and_provenance.md`.

year near a fixed annual anchor date and then frozen in a crawl map.

Several software choices are methodologically consequential because they affect corpus membership. WET and WARC records are parsed with `warcio`, WARC pointers are resolved through a local `pywb`-based index server, archived HTML is converted to main text with Trafilatura and Resiliparse, publication dates are recovered with the `htmldate` library, and post-extraction filtering uses DataTrove quality filters followed by English-language filtering with `py3langid` (Bevendorff et al., 2018; Barbaresi, 2021; Penedo et al., 2026; Lui and Baldwin, 2012; Barbaresi, 2020).[2]

The data collection spans 13 annual Common Crawl snapshots from 2014 to 2026. This window maximises temporal depth while remaining compatible with a stable WET-first, WARC-second workflow: 2014 is the first year with WET/WARC-format crawls large enough for efficient plaintext scanning, denominator construction, and HTML-based validation (earlier archives use older, less comparable formats that require additional handling) and 2026 the latest collection year available to the project.

In the trend track, the pipeline scanned 55.6 million web pages and retained 156,189 pages featuring validated term hits. In the corpus track, it scanned 220 million web pages and retained 336,178 analysis-ready documents. Of these, 87,173 documents contain ADHD or autism target terms. Target membership is non-exclusive: 31,354 documents contain ADHD terms, 67,614 contain autism terms, and 11,795 contain both. The collection was run on an AWS `m7i-flex.large` instance, featuring 2 vCPUs, 8 GiB of RAM, and up to 12.5 Gbps network bandwidth (AWS, 2026). On this instance type, corpus throughput was approximately one hour per million scanned WET records.

3.2 Target Terms

Two target concepts were selected for diachronic semantic analysis: ADHD and autism. Target documents were retrieved using the matching expressions shown in Table 3.1. The abbreviation *ASD* (autism spectrum disorder) was retained only when *autism* occurred within $\pm$200 characters, reducing false positives from unrelated acronym use. The acronym *ADD* (attention deficit disorder) was not included as a matching expression because ADHD is the current canonical diagnostic label, whereas ADD is older and colloquial (American Psychiatric Association, 2013); more importantly, *ADD* would create substantial false positives in WET scanning because it overlaps with the common verb *add* and with web chrome such as "add to cart". The broader expression

[2] Both `warcio` and `pywb` are open-source Webrecorder projects; see `https://github.com/webrecorder/warcio` and `https://github.com/webrecorder/pywb`.

`attention[-\s]?deficit` retains coverage of relevant expanded forms while avoiding this high-noise acronym.

Table 3.1: Target-term matching expressions.

Concept	Matching expressions
ADHD	`\badhd\b`; `attention[-\s]?deficit`
Autism	`\bautism\b`; `\bautistic\b`; `autism[-\s]?spectrum`; `\bASD\b`

For comparison, three negative, non-clinical emotion terms were selected with sufficient coverage and interpretable usage: *frustration*, *sadness*, and *loneliness*. These terms are not exact semantic controls for ADHD and autism; they provide a baseline for separating target-specific change from broader shifts in negative affective language in the corpus.

3.3 Preprocessing

Following Baes, Haslam, and Vylomova (2024), preprocessing was organised around analysis-specific corpus representations rather than a single all-purpose text file. The downstream semantic analyses use a shared mention-level context table built from the extraction pipeline's corpus product. Target and baseline mentions were re-detected using the frozen collection patterns (see Table 3.1), including the ASD disambiguation rule, and overlapping matches within the same analysis unit were resolved by keeping the longest span. This prevents expressions such as *autism spectrum* from being counted twice as both a phrase and a shorter nested form. Documents containing both ADHD and autism terms were allowed to contribute to both target groups, with separate mention-level contexts emitted for each concept. Same-sentence acronym-and-expansion pairs occurring within a short local window were also collapsed, so cases such as "attention deficit hyperactivity disorder (ADHD)" contribute one conceptual ADHD context rather than two near-duplicate rows. To limit domination by repetitive documents while preserving repeated-use signal, each document could contribute at most three mentions per analysis unit. The semantic time variable was then defined as publication year and only contexts with parseable publication dates between 2014 and 2026 were retained for the shared LSC table. This yielded 293,670 mention contexts from 212,651 documents and 135,771 registered domains,[3] including 28,611 ADHD

[3]Relative to the 336,178 analysis-ready corpus documents, mention re-detection produced 664,606 raw matches, reduced to 643,735 by overlap resolution, to 633,026 by acronym–expansion collapse, and to 465,400 candidate contexts (from 336,144 documents) by the three-mention cap. The publication-year filter then removed 161,122 contexts published before 2014 and 10,608 without parseable dates. The document-level attrition from 336,178 to 212,651 therefore predominantly reflects archived pages published before the study window, since each yearly crawl also captures older pages.

contexts, 68,253 autism contexts, and separate baseline series for *frustration*, *sadness*, and *loneliness*. For each retained mention, the table stores the matched form, raw-form collapse diagnostics, registered domain, publication and crawl metadata, a $\pm$5-token window for affective collocate analyses, and target-sentence passages for breadth and thematic analyses.

3.4 NRC–VAD Lexicon

The affective analyses use the NRC Valence, Arousal, and Dominance (VAD) Lexicon v2.1 (Mohammad, 2025), rather than the Warriner norms used in SIBling (Baes, Haslam, and Vylomova, 2024). NRC–VAD offers several advantages for the present study. It is substantially larger, providing human ratings for approximately 45,000 English words and 10,000 multi-word phrases, compared with 13,915 words and no phrase entries in Warriner norms. It has also been reported to have higher reliability, with an aggregate reliability estimate of 0.923 across the three VAD dimensions, compared with 0.823 for the Warriner norms.

NRC–VAD scores range from -1 to 1. Higher valence scores indicate more positive affect, higher arousal scores indicate greater emotional activation, and higher dominance scores indicate greater perceived control or power. This study uses valence to estimate whether the local contexts of target terms become more positive or negative over time, and arousal to estimate changes in emotional intensity. Dominance is included in the source lexicon but is not used in the present analyses. The ratings were produced using Best–Worst Scaling, in which annotators are shown four items and asked to identify the item that best and worst represents the property being rated (Mohammad, 2025).

4 | Methods

The analysis adapts Baes et al.'s 2024 SIBling framework, which characterises lexical semantic change (LSC) along interpretable dimensions rather than treating change as a single aggregate distance. The study estimates annual trajectories for ADHD and autism discourse across salience, frame composition, intensity, breadth, sentiment, and neighbour similarity evolution. Salience measures how often target and baseline terms occur in sampled Common Crawl slices; frame composition tracks the balance between clinical/disorder-construct and lived-experience uses of ADHD and autism; intensity and sentiment measure the affective arousal and valence of local collocates, respectively; breadth measures contextual dispersion among target-aware embeddings; and neighbour similarity evolution tracks how the semantic proximity between each target concept and its recurrent neighbouring terms changes across the study period. Post-hoc diagnostics inspect which VAD-matched collocates and high-distance context words contribute most to the sentiment, intensity, and breadth results. Salience is indexed by Common Crawl source year, while frame composition and all semantic analyses use document publication year as the annual time axis. For ADHD and autism, semantic estimates are reported overall and separately within clinical/disorder and lived-experience frames; baseline terms are not frame-labelled, because the clinical/lived-experience distinction is specific to ADHD and autism discourse. [1]

4.1 Salience

Salience estimates whether ADHD and autism terms become more or less frequent in sampled Common Crawl web discourse over time. Unlike the semantic analyses, which use document publication year, salience is measured on the Common Crawl source-year axis. This choice is technical rather than conceptual: the denominator must cover the full yearly WET sample entering term matching, including pages without target hits, and publication dates are recovered only downstream, for WARC-

[1] The full code implementation of all analyses is available at `https://github.com/jako6f/msc-nlp-therapy-speak/tree/main/notebooks`.

extracted hit documents. A publication-year denominator would therefore be unavailable for the non-hit background corpus. Source year is not identical to publication year (Baack, 2024), so salience should be interpreted as source-year prominence in the sampled crawl rather than as a direct estimate of publication-year prevalence.

For each analysis unit u and Common Crawl source year Y, the numerator is the number of WARC-validated term hits, and the denominator is the number of tokens in minimum-length WET records entering the annual scan. Reported salience rates are scaled to hits per million WET tokens:

$$\text{Salience}_{u,Y} = 1{,}000{,}000 \times \frac{H_{u,Y}^{\text{WARC}}}{T_{Y}^{\text{WET}}}. \tag{1}$$

4.2 Frame Classification

Frame classification precedes the semantic analyses because target-term contexts may change not only in meaning but also in discourse composition. In web text, ADHD and autism may be framed as diagnoses, disorder constructs, service categories, identities, lived experiences, community labels, or incidental boilerplate. Treating all target contexts as one semantic population would therefore risk conflating LSC with shifts in the prevalence of these frames. This concern follows Pisl, Bucur, and Podina (2025), who show that apparent semantic-intensity trends can be explained by the changing mental-health context in which a term appears.

The annotation unit is the target sentence plus adjacent sentence context. Each ADHD/autism passage is labelled hierarchically. First, the passage is coded for whether it contains substantive target discourse: passages that are thin, list-like, navigational, promotional, generic, incidental, noisy, or otherwise insufficient for target-specific interpretation are assigned to the non-substantive or insufficient category. Substantive passages are then coded on two non-exclusive axes: whether clinical framing is present and whether lived-experience framing is present. Clinical framing covers diagnosis, disorder status, symptoms, impairment, treatment, services, medication, research, epidemiology, DSM/ICD-style categories, and educational or clinical support needs. Lived-experience framing covers identity, self-understanding, family or first-person experience, neurodivergent community, stigma, accommodation, everyday coping, belonging, pride, and embodied or social experience. The two frame axes are converted deterministically into five derived strata: substantive passages with clinical but not lived-experience framing are labelled `clinical-only`; passages with lived-experience but not clinical framing are labelled `lived-only`; passages with both are labelled `mixed`; substantive passages with neither are labelled `substantive-other`; and non-substantive or insufficient passages form the fifth stratum.[2]

Frame labels were generated with the assistance of large language models (LLMs), organised as an Annotation with Critical Thinking (ACT) workflow adapted from Lin et al. (2025). Rather than treating LLM output as final annotations, ACT uses one LLM as an annotator, a second as a critic that estimates which annotations are most likely to be erroneous, and human adjudication to resolve uncertain or high-risk cases. First, a 200-passage human pilot was used to refine the codebook. The locked codebook was then applied to 3,000 ADHD/autism passages stratified by year and target using OpenAI's Codex GPT-5.5 with high reasoning effort, producing initial machine annotations for the full annotation batch. Gemini 3 Flash Preview, accessed via the Gemini CLI, then reviewed each Codex-generated annotation and assigned an error-risk score. Following ACT, human review focused on cases exceeding a predefined error-risk threshold, resulting in the manual review and correction of 400 cases. This procedure preserved human control over difficult boundary cases while reducing the amount of fully manual annotation required. To guard against critic blind spots, a random sample of lower-risk cases was also inspected.

The corrected labels were used to train a hierarchical classifier over `all-MPNET-base-v2` passage embeddings. The classifier uses three balanced logistic-regression heads with standardised features: one head predicts substantive target discourse for all labelled examples, while the clinical and lived-experience heads are trained only on substantive examples. Year, URL, and domain metadata are excluded from the classifier features to reduce leakage from temporal or source-specific artefacts. A separate 200-passage human validation set was held out from codebook development, LLM annotation, criticism, correction, and model training.[3] After validation, the classifier was applied to all ADHD/autism contexts, producing hard frame labels and frame probabilities for downstream analysis.

4.3 Sentiment

Sentiment captures whether the local connotational environment of a target term becomes more positive or more negative over time. Following the collocate-based logic of Baes, Haslam, and Vylomova (2024), the measure is computed from words and phrases occurring in a $\pm$5-token window around each target or baseline mention. Un-

[2]The full frame codebook is available at `https://github.com/jako6f/msc-nlp-therapy-speak/tree/main/notebooks/01_classification/codebooks`.

[3]Both hand-coded rounds—the 200-passage pilot and this 200-passage validation set—are available at `https://github.com/jako6f/msc-nlp-therapy-speak/tree/main/data/interim/lsc/classification/human_labels`, where the `annotation_round` column distinguishes them. The two sets are disjoint, and neither overlaps the 3,000 LLM-annotated passages.

like that study, the present measure uses NRC–VAD v2.1 valence scores rather than Warriner norms, because NRC–VAD provides broader contemporary English coverage and includes multi-word expressions (Mohammad, 2025).

For each mention, the focal lexical material itself is removed from the scoring window. The remaining context is tokenised, part-of-speech tagged, and lemmatised with spaCy `en_core_web_sm`. Following the same preprocessing logic, uninformative tokens—punctuation, symbols, spaces, particles, numerals, non-alphabetic material, one-character lemmas, and spaCy stopwords—are excluded before scoring. NRC–VAD entries are normalised with the same lemmatisation and filtering procedure; multi-word lexicon entries are retained only when all component tokens pass the collocate filter. Multi-word expressions are matched greedily before unmatched unigram tokens; if several surface entries collapse to the same lemma phrase, their VAD scores are averaged.

For analysis unit u, publication year Y, and reported stratum s, annual sentiment is

$$\text{Sentiment}_{u,Y,s} = \frac{\sum_{w \in C_{u,Y,s}} f_{w,u,Y,s} V(w)}{\sum_{w \in C_{u,Y,s}} f_{w,u,Y,s}}, \tag{2}$$

where $C_{u,Y,s}$ is the set of NRC–VAD-matched collocates in the local windows, $f_{w,u,Y,s}$ is the frequency of collocate w, and $V(w)$ is its NRC–VAD valence score. A post-hoc contributor diagnostic groups target-frame observations into early, middle, and late periods and ranks collocates by their frequency-weighted positive and negative valence contributions.

4.4 Intensity

Intensity operationalises vertical concept creep as the decline in the affective arousal of local target contexts. The measure uses the same local collocates as the sentiment analysis (see above) and differs only in the VAD score being averaged. Consequently, all preprocessing, target-term exclusion, lemmatisation, multi-word matching, frame stratification, and coverage reporting are inherited.

Annual intensity for analysis unit u, publication year Y, and reported stratum s is

$$\text{Intensity}_{u,Y,s} = \frac{\sum_{w \in C_{u,Y,s}} f_{w,u,Y,s} A(w)}{\sum_{w \in C_{u,Y,s}} f_{w,u,Y,s}}, \tag{3}$$

where $A(w)$ is the NRC–VAD arousal score for collocate w. The intensity post-hoc diagnostic applies the same period grouping and contributor-ranking procedure to NRC–VAD arousal scores.

4.5 Breadth

Breadth operationalises horizontal concept creep as contextual dispersion: the more diverse the contexts in which a target term appears, the higher its breadth score. Baes, Haslam, and Vylomova (2024) estimate breadth using *sentence-level* contextual embeddings. This study replaces that generic sentence-embedding representation with XL-LEXEME, a *target-aware word-in-context* (WiC) model designed for LSC detection (Cassotti et al., 2023), so that ADHD, autism, and the baseline terms are represented as target uses rather than as undifferentiated sentence topics.

Each candidate context is marked with explicit XL-LEXEME target delimiters (e.g., "Many `<t>` autistic `</t>` adults describe masking in workplace settings.") The target sentence is used by default; if it is too short, the sentence-plus-adjacent context is used instead. Identical marked contexts are encoded once and reused through an embedding index. For each analysis unit, year, and frame stratum, XL-LEXEME produces a contextual embedding of the marked target use. These embeddings are L2-normalised, and breadth is then calculated as the mean pairwise cosine distance among all target-use embeddings in the corresponding cell.

For analysis unit u, publication year Y, and reported stratum s, with $N_{u,Y,s}$ contextual embeddings $\mathbf{v}_1, \ldots, \mathbf{v}_{N_{u,Y,s}}$, breadth is

$$\text{Breadth}_{u,Y,s} = \frac{2}{N_{u,Y,s}(N_{u,Y,s} - 1)} \sum_{i=1}^{N_{u,Y,s}-1} \sum_{j=i+1}^{N_{u,Y,s}} \left(1 - \frac{\mathbf{v}_i \cdot \mathbf{v}_j}{\|\mathbf{v}_i\| \|\mathbf{v}_j\|}\right). \tag{4}$$

Higher values indicate greater average dissimilarity among target uses in that year and stratum. The implementation computes this mean in closed form over the L2-normalised embedding vectors, avoiding materialising the full pairwise distance matrix in memory. The post-hoc breadth diagnostic ranks contexts by their average cosine distance to the other contexts in the same target-frame-period cell and summarises frequent content lemmas among the highest-distance contexts.

4.6 Neighbour Similarity Evolution

Baes, Haslam, and Vylomova (2024) operationalise thematic content using a top-down pathologisation dictionary, which is appropriate for terms that can refer either to ordinary affective states or to clinical constructs, such as *anxiety* and *depression*. Because

ADHD and autism are diagnostic concepts by definition, the present study instead estimates bottom-up target-neighbour trajectories, following the pairwise similarity time-series of type-level embeddings in Vylomova and Haslam (2021). This analysis asks which content words become more or less distributionally close to each target concept across publication years.

Models are estimated separately for ADHD and autism, stratified by frame. Baseline terms are not modelled, because neighbour similarity evolution is used to characterise target-specific thematic associations rather than to produce a scalar target–baseline comparison. The modelling input is the target-centred passage, using document publication year as the time axis. Raw target forms are canonicalised before modelling so that the type-level embedding represents the target concept rather than individual spellings or abbreviations (i.e., ADHD variants are mapped to `adhd_concept`, and autism variants are mapped to `autism_concept`). Passages are lemmatised with spaCy `en_core_web_sm` and filtered to retain content words and canonical concept tokens, while removing punctuation, numerals, stopwords, one- and two-character tokens, and common web-boilerplate tokens. Contexts with fewer than five retained tokens, or with no canonical target token, are excluded.

For each target-frame corpus, a global skip-gram Word2Vec model is trained with 200-dimensional vectors, a context window of 10 tokens, a minimum corpus count of 5, and 10 training epochs, following Vylomova and Haslam (2021). Annual models are then initialised from the corresponding global model and further trained for 10 epochs on passages from each publication year. This gives the annual models a shared target-frame starting space while still allowing yearly neighbour associations to vary.

For target concept w_c, candidate neighbour w_j, target unit u, frame stratum f, and publication year t, the neighbour-similarity score is defined as the cosine similarity between the annual embedding of the canonical target concept and the annual embedding of the candidate neighbour:

$$\mathrm{NSE}_{u,f,t}(w_j) = \cos\left(\mathbf{v}_{w_c}^{(u,f,t)}, \mathbf{v}_{w_j}^{(u,f,t)}\right) = \frac{\mathbf{v}_{w_c}^{(u,f,t)} \cdot \mathbf{v}_{w_j}^{(u,f,t)}}{\|\mathbf{v}_{w_c}^{(u,f,t)}\| \|\mathbf{v}_{w_j}^{(u,f,t)}\|}. \tag{5}$$

Annual neighbours are extracted only when the canonical target token occurs at least 20 times in the relevant target-frame-year corpus. Candidate neighbours must occur at least five times in the same annual corpus, and the five most similar eligible neighbours are retained for each model. To reduce noise in the resulting trajectories, a neighbour must appear in the annual top-five list in at least two years and have finite similarity estimates in at least ten of the thirteen publication years.

4.7 Statistical Analysis

Annual estimates are the primary objects of interpretation. For all semantic analyses, uncertainty intervals are estimated by document-level bootstrap resampling within each analysis-unit, year, and frame-stratum cell, using 500 bootstrap repetitions and the 2.5th and 97.5th percentiles of the bootstrap distribution. The document is the resampling unit because multiple mentions and collocates from the same document are not independent. Reported results refer to two-tailed tests with no correction for multiple testing.

Residual autocorrelation is assessed using a Durbin–Watson-style statistic computed on the ordinary least-squares (OLS) residuals e_t:

$$D = \frac{\sum_{t=2}^{T}(e_t - e_{t-1})^2}{\sum_{t=1}^{T} e_t^2}. \tag{6}$$

Annual series are flagged when $D < 1.25$ or $D > 2.75$. When a scalar annual series is flagged, a first-order autoregressive (AR(1)) model is estimated as a sensitivity check and reported in the appendix; otherwise, the OLS slope remains the main summary (Chatterjee and Hadi, 2012). Quadratic fits are computed only for annual series whose linear-model residual diagnostics indicate autocorrelation, and they are treated as diagnostic checks rather than primary trend estimates.

Two caveats apply. First, the present study reports 34 uncorrected OLS trend tests (including Appendix), so roughly 1.7 nominally significant slopes are expected by chance at $\alpha = .05$, and isolated significant secondary slopes should be read accordingly. Second, with 13 annual observations, a slope reaches $p < .05$ only when $|B| \geq 2.20 \times SE$ ($t_{11;.975}$). For the four directional hypothesis tests (overall intensity and breadth, for each target), the fitted standard errors imply that the smallest 2014–2026 changes that would have reached significance are about .013 (ADHD) and .022 (autism) on the -1 to 1 arousal index, and .008 and .006 in mean pairwise cosine distance. Null results rule out larger linear changes, not smaller drifts.

5 | Results

This chapter reports diachronic results for salience, frame composition, sentiment, intensity, breadth, and thematic neighbour similarity for ADHD and autism. Salience is indexed by Common Crawl source year, while the semantic measures use document publication year. Annual estimates and their bootstrap intervals show the observed trajectories; ordinary least-squares (OLS) models summarise net linear movement over the 13 annual observations. In these models, B is the unstandardised change per year and adjusted R^2 summarises model fit. The reported p-values are descriptive and uncorrected.

The semantic results foreground the overall aggregate—which combines clinical-only, lived-experience-only, and mixed contexts—and the two clear, mutually exclusive frames: clinical-only and lived-experience-only. Mixed contexts are not plotted separately: this smaller stratum generally tracked between the overall and clinical trajectories and added visual overlap. Substantive-other contexts were sparse and outside the main frame contrast; non-substantive or insufficient contexts were treated as a quality and composition category rather than a semantic frame.

Table 5.1 reports the target trend models. At the uncorrected $p < .05$ level, autism salience declined; frame composition shifted toward lived experience; no target sentiment OLS slope differed from zero; intensity increased only in autism clinical contexts; and breadth decreased for ADHD overall, ADHD clinical contexts, and autism lived-experience contexts. The autism clinical breadth OLS slope was nominally positive but was not retained by the AR(1) sensitivity model.

Table 5.1: Descriptive annual trend models for ADHD and Autism LSC trajectories by frame.

Measure	Target	Frame	$B(SE)$	β	Adj. R^2
Salience	ADHD	Overall	0.0073 (0.0043)	0.46	0.14
Salience	Autism	Overall	-0.0228^{*} (0.0087)	-0.62	0.33
Sentiment	ADHD	Overall	$0.0023^{\dagger}$ (0.0014)	0.46	0.14
Sentiment	ADHD	Clinical	$0.0013^{\dagger}$ (0.0015)	0.24	-0.03
Sentiment	ADHD	Lived experience	0.0002 (0.0017)	0.03	-0.09
Sentiment	Autism	Overall	0.0012 (0.0006)	0.50	0.18
Sentiment	Autism	Clinical	$0.0022^{\dagger}$ (0.0015)	0.42	0.10
Sentiment	Autism	Lived experience	-0.0038 (0.0022)	-0.46	0.14
Intensity	ADHD	Overall	0.0001 (0.0005)	0.04	-0.09
Intensity	ADHD	Clinical	0.0002 (0.0006)	0.10	-0.08
Intensity	ADHD	Lived experience	0.0015 (0.0009)	0.45	0.13
Intensity	Autism	Overall	0.0011 (0.0008)	0.36	0.05
Intensity	Autism	Clinical	0.0010^{*} (0.0004)	0.56	0.25
Intensity	Autism	Lived experience	0.0025 (0.0021)	0.34	0.04
Breadth	ADHD	Overall	-0.0020^{***} (0.0003)	-0.89	0.77
Breadth	ADHD	Clinical	-0.0018^{***} (0.0003)	-0.85	0.69
Breadth	ADHD	Lived experience	$-0.0004^{\dagger}$ (0.0006)	-0.18	-0.06
Breadth	Autism	Overall	$0.0001^{\dagger}$ (0.0002)	0.14	-0.07
Breadth	Autism	Clinical	$0.0007^{*\dagger}$ (0.0003)	0.58	0.28
Breadth	Autism	Lived experience	-0.0010^{*} (0.0004)	-0.60	0.30

Note. Cells report annual unstandardised OLS slopes as $B(SE)$; β is the standardised year coefficient. Salience uses Common Crawl source year and has only an overall target row; semantic measures use document publication year. Significance markers: $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$. † indicates a residual-autocorrelation flag; AR(1) sensitivity estimates for flagged rows are reported in Appendix Table A.4. P values are descriptive and uncorrected.

5.1 Salience

Figure 5.1 presents WARC-validated source-year salience. Autism remained more frequent than ADHD throughout 2014–2026, but its fitted trajectory declined by 28.9% from 2014 to 2026 ($B = -0.0228$, $p = .024$, adjusted $R^2 = .33$). ADHD moved in the opposite direction, with a fitted 27.0% increase over the same period, although the annual slope did not differ from zero ($B = 0.0073$, $p = .118$, adjusted $R^2 = .14$). The indexed panel shows the target trajectories relative to the comparators; comparator trend models are reported in Appendix Table A.3.

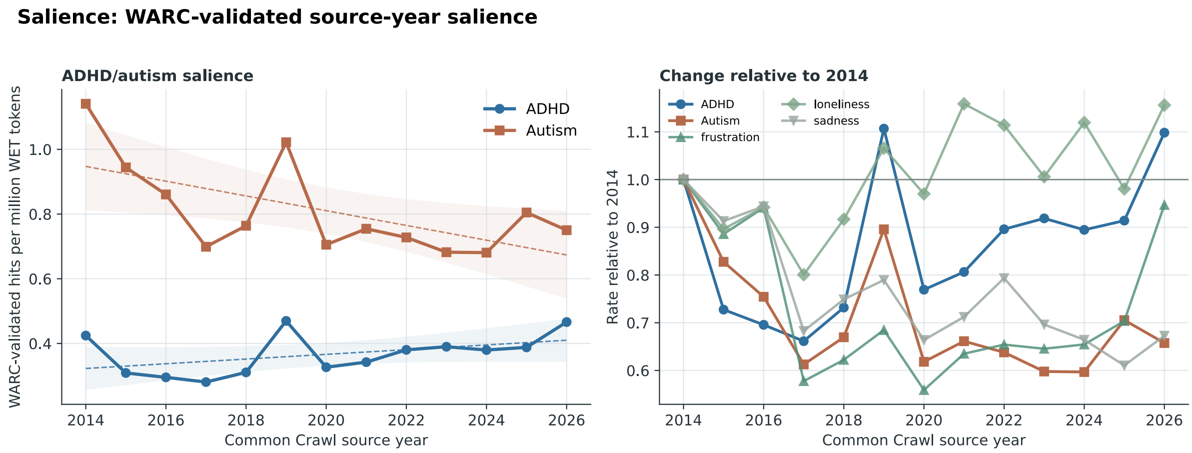

Figure 5.1: WARC-validated ADHD and autism hits per million WET tokens by Common Crawl source year. The right panel indexes each target and comparator series to its 2014 rate.

5.2 Frame Classification

The hierarchical classifier assigned frame labels to all 96,864 target contexts: 28,611 for ADHD and 68,253 for autism. Table 5.2 shows that it performed well on held-out passages, with the strongest performance for substantive-context detection and clinical framing and weaker, though still usable, performance for lived-experience framing.

Table 5.2: Held-out validation performance for the hierarchical frame classifier.

Validation target	Support	Precision	Recall	F1	Accuracy	Bal. acc.
Substantive target discourse	141/200	.887	.837	.861	.810	.791
Clinical framing	105/141	.903	.886	.894	.844	.804
Lived-experience framing	46/141	.704	.826	.760	.830	.829

Note. Support gives positive cases over the validation denominator. Substantive target discourse is evaluated on all 200 held-out passages. Clinical and lived-experience performance is evaluated among the 141 passages labelled as substantive target discourse by the human coder.

Of the labelled contexts, clinical-only comprised 42.1% of ADHD and 29.8% of autism contexts; lived-experience-only comprised 12.6% and 18.8%, respectively. The remaining assignments were mixed (7.0% and 9.3%), non-substantive or insufficient (35.8% and 38.8%), and substantive-other (2.5% and 3.3%). Figure 5.2 shows a shift toward lived-experience framing, especially in ADHD discourse. In the full predicted composition (Panel A), lived-experience-only assignments rose from 10.7% to 15.7% of ADHD contexts between 2014–2017 and 2022–2026, while clinical-only assignments fell from 44.4% to 39.1%; autism moved in the same direction more weakly. The annual trend model then restricts the denominator to the clear clinical-only and lived-experience-only assignments, estimating the lived-experience share among those two frames (Panel B). On this contrast, ADHD increased by 1.07 percentage points per year ($p < .001$), a fitted 12.8-point rise from 2014 to 2026; autism increased by 0.59 points per year ($p = .013$), a fitted 7.0-point rise, although this series was flagged for residual autocorrelation and the AR(1) sensitivity slope did not differ from zero ($p = .370$). The trend models are reported in Appendix Table A.1. A post-hoc temporal-stability audit (Appendix Table A.2) found no evidence that this rise reflects drifting classifier error: although clinical recall was nominally lower in the late period, lived-experience error rates were homogeneous, with precision highest in the late period, and the human-adjudicated labels reproduced the trend independently of the classifier (+1.11 percentage points per year, bootstrap 95% CI [0.40, 1.88]).

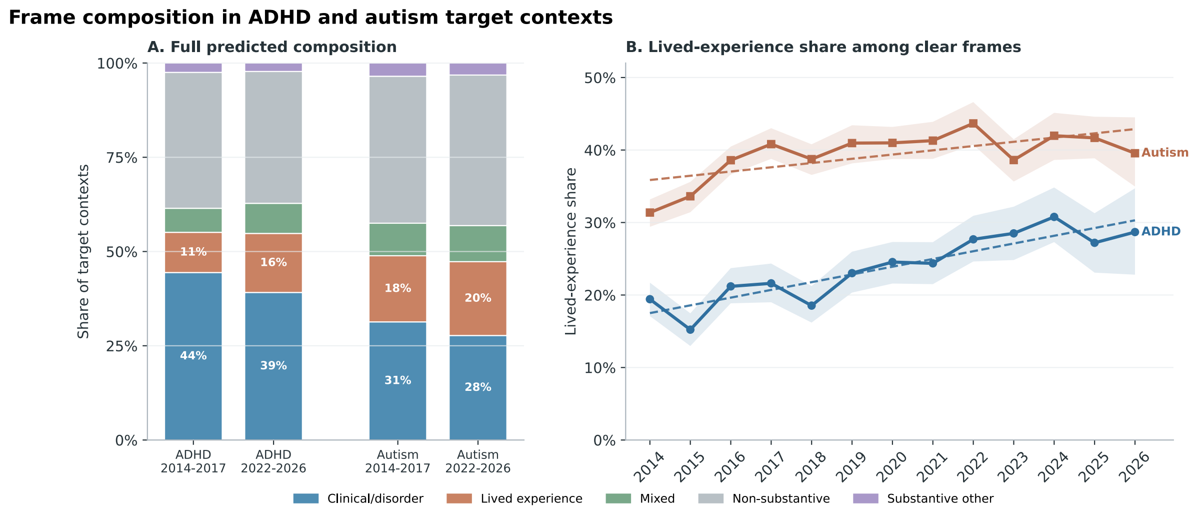

Figure 5.2: Predicted frame composition and clear-frame lived-experience trajectories for ADHD and autism target contexts. Panel A retains all predicted derived-frame labels and compares early (2014–2017) with late (2022–2026) periods. Panel B shows annual lived-experience share among contexts assigned either clinical-only or lived-experience-only; shaded bands are 95% document-bootstrap intervals and dashed lines are OLS trend summaries over publication years.

5.3 Sentiment

Sentiment estimates were based on 17,649 ADHD and 39,463 autism contexts in the overall aggregate. At least one NRC–VAD match occurred in 17,255 ADHD contexts (97.8%) and 39,200 autism contexts (99.3%), yielding 58,666 and 144,668 matched collocate occurrences, respectively. Within individual years, at least 96.2% of ADHD contexts and 98.9% of autism contexts contributed a match, while matched-token coverage was at least 88.7% and 87.6%, respectively. The resulting annual valence index ranges from -1 to 1, with higher values indicating more positive local wording.

ADHD. Annual overall valence ranged from .044 to .106 (M = .076, SD = .020). Its positive linear slope did not differ from zero ($B = 0.0023$, $p = .113$, adjusted $R^2 = .14$); neither did the clinical or lived-experience slopes. The overall and clinical residuals were flagged for autocorrelation, and the AR(1) sensitivity slopes also did not differ from zero. Quadratic diagnostics fitted U-shaped overall and clinical trajectories, with minima in Q3 2018 and Q2 2019 and adjusted-R^2 gains of .29 and .60 over the corresponding linear models.

Autism. Annual overall valence ranged from .137 to .173 (M = .149, SD = .009). Neither the overall slope ($B = 0.0012$, $p = .081$, adjusted $R^2 = .18$) nor the clinical and lived-experience OLS slopes differed from zero. The clinical residuals were flagged for autocorrelation; the AR(1) sensitivity slope was positive ($B = 0.0052$, $p = .040$). The quadratic model fitted a U-shaped clinical trajectory with a minimum in Q1 2019 and an adjusted-R^2 gain of .65.

Post-hoc contributor patterns indicate that positive valence contributions were concentrated around child- and family-related vocabulary, while lower-valence contributions came mainly from diagnostic, disability, and co-occurring-condition terms, including *autism* in ADHD contexts and *adhd* in autism contexts (Appendix Table A.6.1).

Figure 5.3 presents the flagged sentiment trajectories with linear and quadratic fits. Coefficients for the quadratic models are reported in Appendix Table A.5.

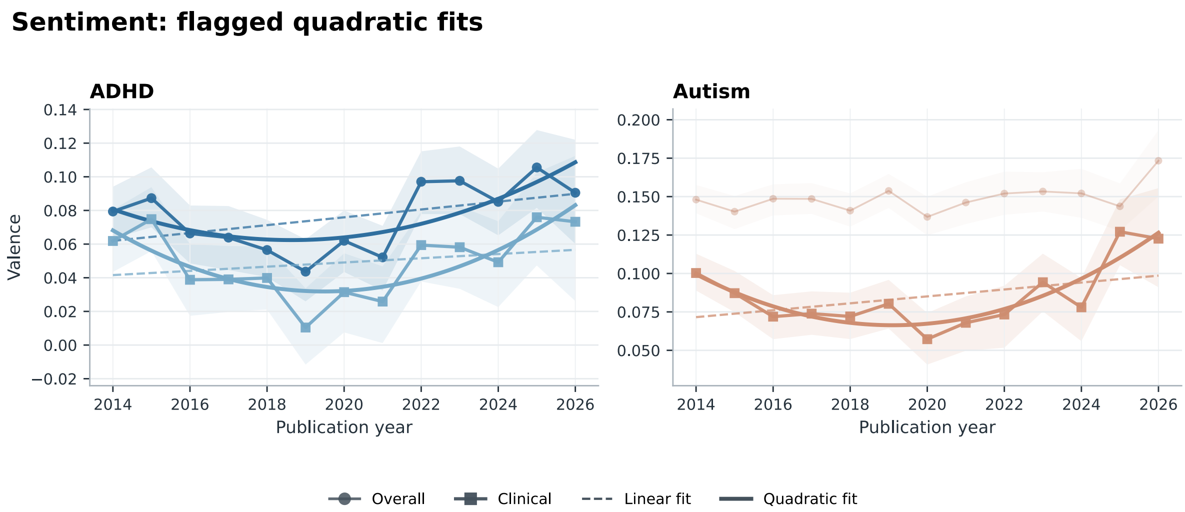

Figure 5.3: Annual mean valence for the flagged ADHD overall, ADHD clinical, and autism clinical trajectories, with 95% document-bootstrap intervals and linear and quadratic fits. Autism overall estimates and intervals are shown as muted contextual reference.

5.4 Intensity

Intensity uses the same NRC–VAD matches as sentiment and is therefore based on the same 17,649 ADHD and 39,463 autism contexts and 58,666 and 144,668 matched collocate occurrences. The context- and token-coverage rates reported above also apply to the annual arousal estimates. The arousal index ranges from -1 to 1, with higher values indicating greater emotional activation.

ADHD. Annual overall arousal ranged from .011 to .032 ($M = .023$, $SD = .007$). Contrary to hypothesis, the fitted slope did not differ from zero overall ($B = 0.0001$, $p = .900$, adjusted $R^2 = -.09$), in clinical contexts ($B = 0.0002$, $p = .738$, adjusted $R^2 = -.08$), or in lived-experience contexts ($B = 0.0015$, $p = .123$, adjusted $R^2 = .13$).

Autism. Annual overall arousal ranged from approximately .000 to .047 ($M = .010$, $SD = .012$). Contrary to expectation, the overall slope was positive, though it did not differ from zero ($B = 0.0011$, $p = .223$, adjusted $R^2 = .05$). Clinical arousal increased nominally ($B = 0.0010$, $p = .048$, adjusted $R^2 = .25$), while the lived-experience slope did not differ from zero ($B = 0.0025$, $p = .252$, adjusted $R^2 = .04$).

Post-hoc arousal contributor patterns were concentrated in diagnostic and difficulty-related vocabulary. ADHD arousal was raised most consistently by terms such as *disorder*, *anxiety*, *hyperactivity*, *challenge*, and *struggle*; autism arousal was raised by *disorder*, *developmental*, *spectrum*, and, in later lived-experience contexts, terms such as *dangerous* and *obsession*. Lower-arousal contributors included treatment-, individual-, people-, and family-related vocabulary (Appendix Table A.6.2). Figure 5.4 presents the target and comparator trajectories. Among the comparators, *sadness* increased; the OLS slopes for *frustration* and *loneliness* did not differ from zero.

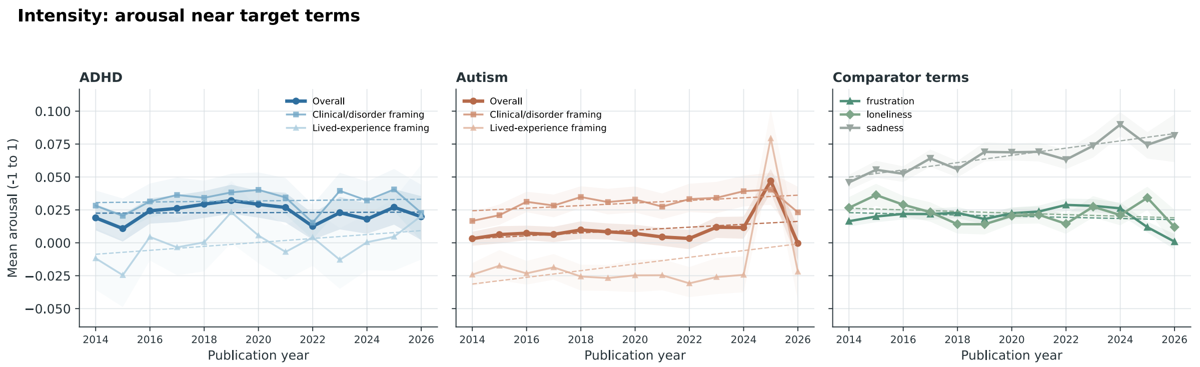

Figure 5.4: Annual mean NRC–VAD arousal for ADHD, autism, and comparator contexts. Shaded bands are 95% document-bootstrap intervals; dashed lines are OLS trend summaries.

5.5 Breadth

Breadth estimates were available for 57,112 target contexts in the three substantive frames. After within-cell exact-duplicate removal, 53,544 contexts contributed to the frame-specific estimates; the same material was also aggregated into the overall target series, yielding 107,081 target analysis rows. The comparator sample contributed 38,349 rows. The annual index is the mean pairwise cosine distance, with higher values indicating greater contextual dispersion.

ADHD. Breadth was .138 in 2014 and .117 in 2026 overall (M = .127, SD = .009). Contrary to hypothesis, it decreased overall ($B = -0.0020$, $p < .001$, adjusted R^2 = .77) and in clinical contexts ($B = -0.0018$, $p < .001$, adjusted R^2 = .69). The lived-experience slope did not differ from zero ($B = -0.0004$, $p = .560$, adjusted $R^2 = -.06$), and its residuals were flagged for autocorrelation. Figure 5.5 presents these trajectories alongside the autism and comparator series.

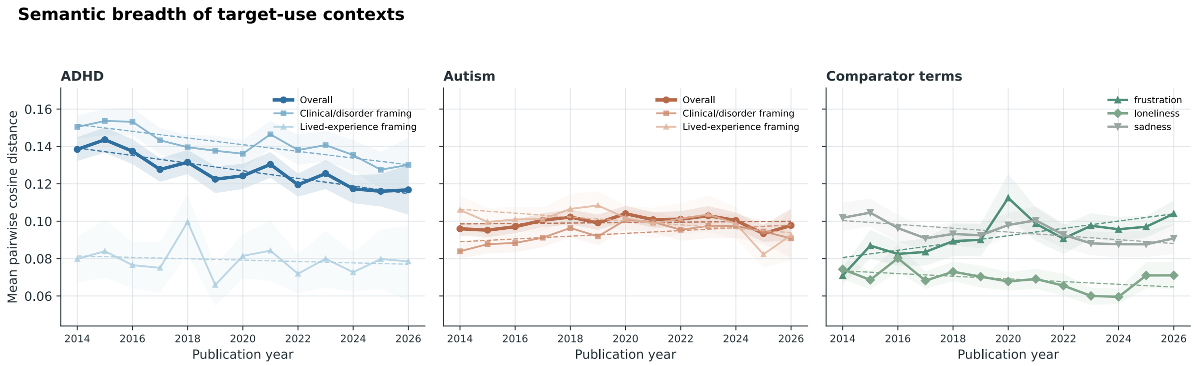

Figure 5.5: Annual mean pairwise cosine distance among target-aware contextual embeddings for ADHD, autism, and comparator terms. Higher values indicate greater contextual breadth; shaded bands are 95% document-bootstrap intervals.

Autism. Breadth was .096 in 2014 and .098 in 2026 overall ($M = .099$, $SD = .003$). Contrary to expectation, its linear slope did not differ from zero ($B = 0.0001$, $p = .637$, adjusted $R^2 = -.07$), and the flagged AR(1) sensitivity slope also did not differ from zero ($p = .753$). Lived-experience breadth decreased ($B = -0.0010$, $p = .030$, adjusted $R^2 = .30$). The clinical OLS slope was nominally positive ($B = 0.0007$, $p = .036$, adjusted $R^2 = .28$), but its residuals were flagged for autocorrelation and the AR(1) sensitivity slope did not differ from zero ($p = .755$).

The breadth diagnostic suggests different lexical profiles among the highest-distance sampled contexts. ADHD high-distance contexts repeatedly surfaced diagnostic, educational, and co-occurring-condition vocabulary, including *disorder*, *impulsivity*, *inattention*, *learning*, *dyslexia*, and *autism*. Autism high-distance contexts were more varied: clinical contexts included terms such as *research*, *organization*, *developmental*, and *treatment*, while overall and lived-experience contexts included broader contextual words such as *world*, *feel*, *people*, and *experience* (Appendix Table A.6.3).

Figure 5.6 presents quadratic fit diagnostics for the two flagged autism series. The overall trajectory had an inverted-U fit with a vertex in Q2 2020, while the clinical fit had a vertex in Q2 2021; the adjusted-R^2 gains over the corresponding linear models were .58 and .51. Appendix Table A.5 reports the model coefficients.

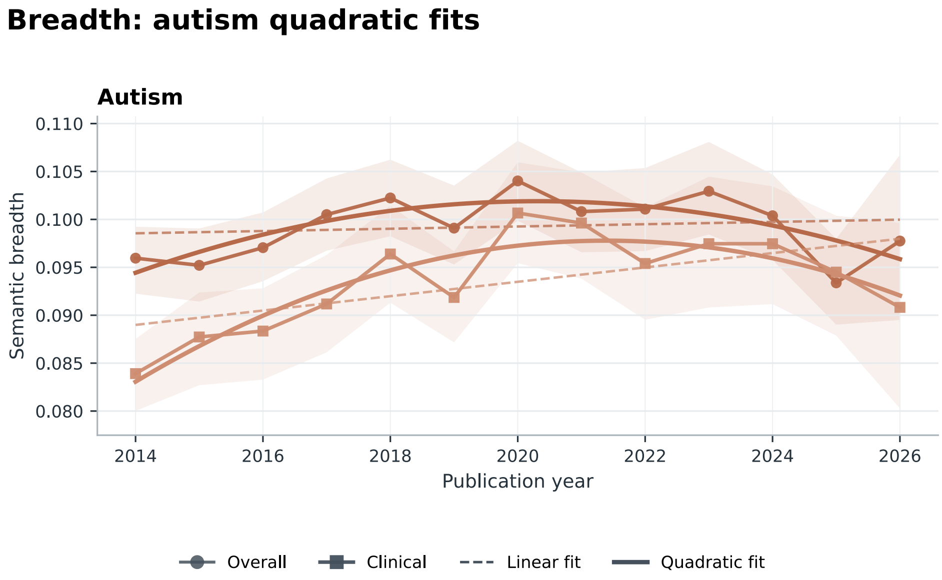

Figure 5.6: Annual overall and clinical autism breadth, with 95% document-bootstrap intervals and linear and quadratic fits. Both series were selected following residual-autocorrelation flags.

5.6 Neighbour Similarity Evolution

The thematic analysis began with 57,112 tokenised target contexts, of which 56,857 remained after content filtering. The complete annual top-five output contained 390 neighbour rows across two targets, three reporting strata, and 13 years. Stability filtering retained 28 neighbours and 364 annual similarity estimates.

For ADHD, the stable overall neighbours were *child*, *disorder*, *symptom*, *diagnose*, and *condition*. The clinical set contained the same diagnostic and child-related vocabulary, while the lived-experience set retained *help*, *child*, and *diagnose*. These patterns are consistent with the corresponding frame definitions. The ADHD trajectories are reported in Appendix Figure A.1.

Figure 5.7 presents the autism trajectories. *Child* remained the highest-similarity stable neighbour overall and in clinical contexts. The remaining clinical neighbours were *disorder*, *research*, *study*, and *developmental*, while the lived-experience set comprised *people*, *child*, *support*, *life*, and *family*—again consistent with the respective frame definitions.

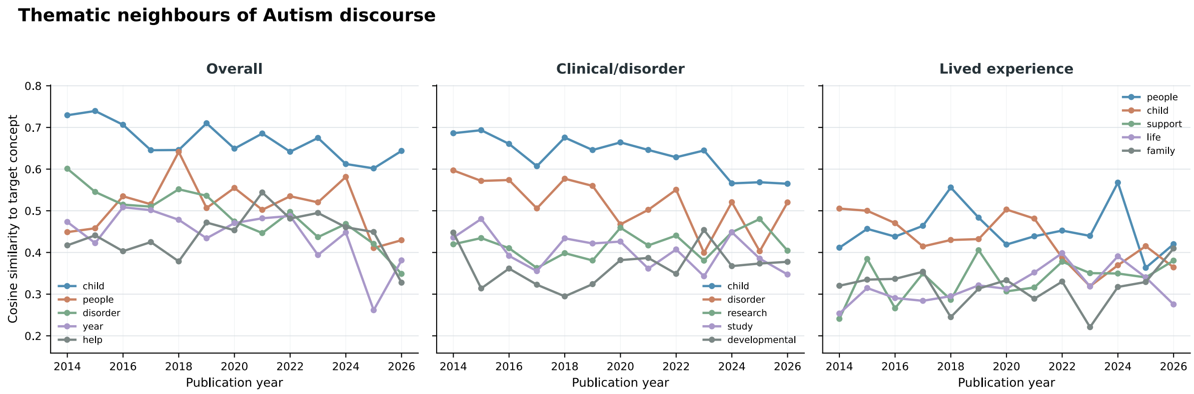

Figure 5.7: Annual cosine similarity between the autism concept token and stable Word2Vec neighbours in overall, clinical/disorder, and lived-experience contexts. Stable neighbours appeared in an annual top-five list in at least two years and had estimates in at least ten years.

5.7 Robustness Checks

The robustness checks compared the main affective and breadth measures with the original SIBling operationalisations: Warriner affect norms for sentiment and intensity, and MPNet sentence embeddings for breadth (Baes, Haslam, and Vylomova, 2024). The annual strata and trend-model convention were held constant.

Under Warriner norms, the ADHD sentiment estimate remained positive but did not differ from zero ($B = 0.0003$, $p = .366$, adjusted $R^2 = -.01$). The autism sentiment estimate was positive ($B = 0.0005$, $p = .018$, adjusted $R^2 = .36$), and its residuals were flagged for autocorrelation; the AR(1) sensitivity estimate also differed from zero ($B = 0.0005$, $p < .001$).

Intensity was more sensitive to the affective resource. The near-zero NRC–VAD ADHD slope ($B = 0.0001$, $p = .900$, adjusted $R^2 = -.09$) became negative under Warriner ($B = -0.0009$, $p < .001$, adjusted $R^2 = .62$), and the AR(1) sensitivity estimate remained negative ($B = -0.0011$, $p = .008$). For autism, the NRC–VAD overall slope was positive but did not differ from zero ($B = 0.0011$, $p = .223$, adjusted $R^2 = .05$), while the Warriner estimate was negative and also did not differ from zero ($B = -0.0002$, $p = .641$, adjusted $R^2 = -.07$).

The breadth trajectories also differed by encoder. ADHD breadth declined under XL-LEXEME ($B = -0.0020$, $p < .001$, adjusted $R^2 = .77$), while the MPNet slope was weaker and did not differ from zero ($B = -0.0004$, $p = .445$, adjusted $R^2 = -.03$). Autism changed from the non-linear, approximately flat overall XL-LEXEME trajectory ($B = 0.0001$, $p = .637$, adjusted $R^2 = -.07$) to a negative MPNet slope ($B = -0.0014$, $p = .001$, adjusted $R^2 = .59$). Figure 5.8 shows the intensity and breadth comparisons as change from each method's 2014 estimate.

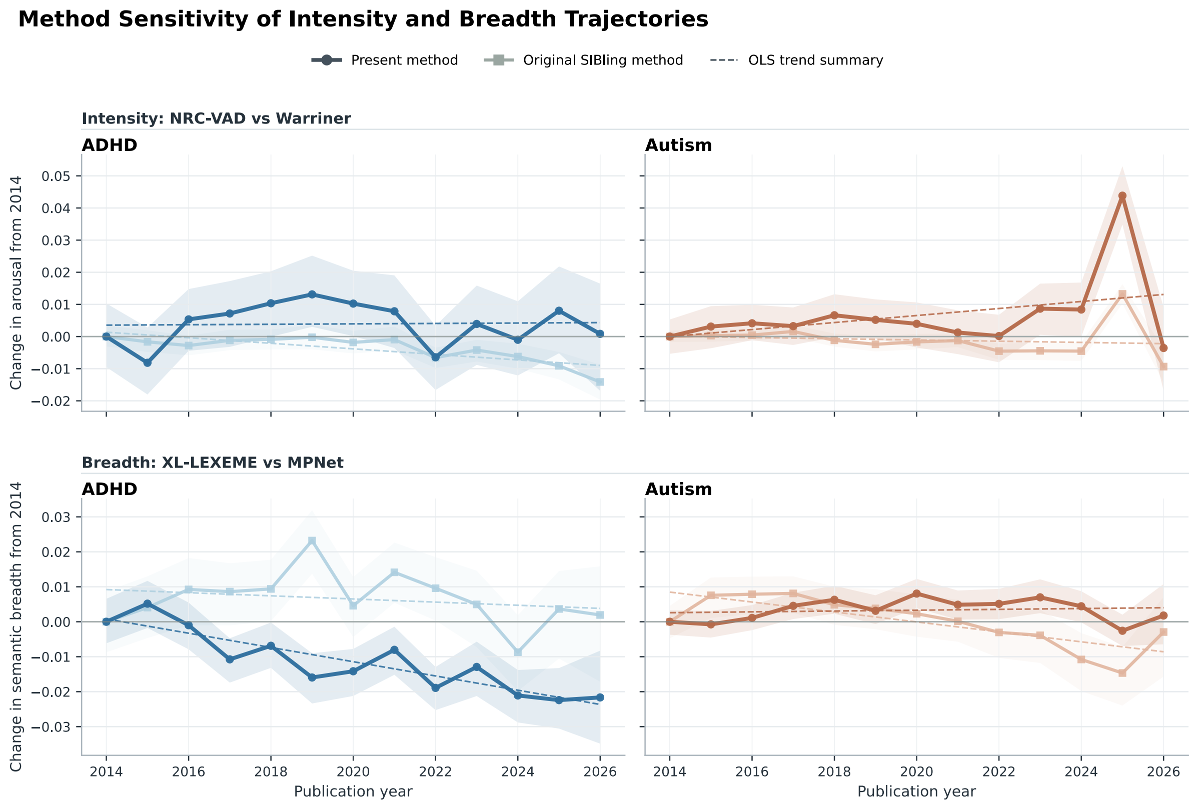

Figure 5.8: Method sensitivity of overall target intensity and breadth trajectories, expressed as change from each method's 2014 estimate. Main measures use NRC–VAD and XL-LEXEME; the original SIBling operationalisations use Warriner norms and MPNet. Shaded bands are shifted annual bootstrap intervals.

6 | Discussion

This study asked whether ADHD and autism—two diagnostic concepts at the centre of contemporary debates about the popularisation of mental health language—have undergone the semantic loosening that concept creep theory predicts, and it did so in general web discourse rather than in the curated corpora on which prior evidence largely rests. The answer, on the measures used here, is that they have not. Across a frame-aware adaptation of the SIBling framework (Baes, Haslam, and Vylomova, 2024), the dominant empirical signature was not milder, broader, or diluted meaning, but a reorganisation of the discourse surrounding two comparatively stable concepts. This chapter first summarises the findings in relation to the research questions (Section 6.1). It then develops three interpretive claims: that the results indicate semantic consolidation rather than concept creep (Section 6.2); that the clearest change in ADHD and autism discourse over the period was compositional—a shift from disorder talk toward lived experience, accompanied by a recent positive valence shift in clinical contexts (Section 6.3); and that the robustness checks expose a measurement sensitivity with implications for the wider literature on lexical semantic change (Section 6.4). Limitations (Section 6.5) and directions for future research (Section 6.6) close the chapter.

6.1 Summary of Findings

With respect to RQ1, ADHD and autism did not become jointly more prominent in sampled general web discourse between 2014 and 2026. Autism salience declined by a fitted 28.9%, while the fitted 27.0% increase for ADHD did not differ from zero at the uncorrected $p < .05$ level. The comparators moved in different directions—*loneliness* rose, *sadness* fell, and *frustration* showed no reliable trend—so the autism decline cannot be attributed to a uniform contraction of negative-affect language in the crawl.

With respect to RQ2, the balance of framing shifted toward lived experience. Among contexts assigned to a single clear frame, the lived-experience share of ADHD discourse rose by 1.07 percentage points per year, a fitted increase from 17.5% to 30.3%; the corresponding autism trend was directionally similar but roughly half the size and was not retained by the AR(1) sensitivity model. The full predicted composition showed the same pattern, with clinical-only shares falling for both targets.

With respect to RQ3, neither of the hypothesised forms of concept creep was supported (Table 5.1). Contrary to the vertical-creep hypothesis, intensity did not decline in any target stratum; the only slope that differed from zero was a nominal increase in autism clinical contexts. Contrary to the horizontal-creep hypothesis, breadth under the target-aware operationalisation decreased for ADHD overall and in ADHD clinical contexts—the strongest semantic trends observed in the study—decreased in autism lived-experience contexts, and was flat for autism overall, with the nominally positive autism clinical slope not surviving the AR(1) check. Sentiment showed no linear trend in any target stratum; instead, the autocorrelation-flagged series traced U-shaped trajectories with minima between mid-2018 and mid-2019, and the AR(1) sensitivity model indicated a positive valence trend in autism clinical contexts.

With respect to RQ4, the thematic content of both discourses changed remarkably little. The 28 neighbours retained by the stability filter were diagnostic and child- or family-centred throughout: *child*, *disorder*, *symptom*, *diagnose*, and *condition* anchored ADHD discourse, while *child* remained the highest-similarity neighbour of autism overall and in clinical contexts, with *people*, *support*, *life*, and *family* characterising lived-experience contexts.

6.2 Semantic Consolidation Rather than Concept Creep

The central substantive finding is a double null with a directional surprise. ADHD and autism did not drift into less emotionally intense contexts, and they did not disperse across a broader range of lexical contexts; where reliable movement occurred, it ran in the opposite direction, most clearly in the sustained contraction of ADHD breadth, under the target-aware operationalisation, from .138 to .117. This pattern is unlikely to reflect measures too blunt to register change: the same instruments detected reliable trends among the comparators, including a broadening of *frustration*, an intensification of *sadness* contexts, and robust positive valence shifts for *frustration* and *loneliness*. The affective and contextual profiles of ADHD and autism were among the more stable series observed.

This stability is intelligible in light of Iacob and Uban's 2026 finding that long-established, codified psychological terms such as *OCD* and *bipolar* showed little year-to-year semantic movement on Reddit, whereas newer vernacular terms such as *gaslighting* shifted substantially. ADHD and autism are institutionally anchored concepts: their reference is stabilised by diagnostic manuals, clinical gatekeeping, service categories, and educational law in ways that ordinary-language harm concepts such as *bullying* or *harassment* (Vylomova and Haslam, 2021) are not. Concept creep in such categories may therefore operate primarily at the level of diagnostic criteria and category application—who is counted as an instance—rather than at the level of the contexts in which the label is used. This reading is consistent with Fabiano and Haslam's 2020 meta-analytic evidence that ADHD's diagnostic criteria inflated substantially across successive DSM revisions, and it suggests a distinction the concept creep literature has tended to elide: a category can expand extensionally, absorbing many more people, while the typical linguistic contexts of its name—which is all that collocate- and embedding-based measures index—remain stable or even converge. On this account, the rising prevalence documented in the epidemiological literature (Zeidan et al., 2022; McKechnie et al., 2023; Cui et al., 2026) need not leave the semantic trace that a naive application of concept creep theory would predict.

The ADHD breadth contraction invites a further interpretation: consolidation of a discourse genre. As ADHD moved from a specialist topic to a mainstream one, its web coverage appears to have converged on a recognisable register—symptom lists, awareness explainers, parenting and educational advice—rather than diversifying. The post-hoc diagnostics support this reading: the highest-distance ADHD contexts remained saturated with diagnostic and educational vocabulary (*disorder*, *impulsivity*, *inattention*, *learning*, *dyslexia*), and the stable neighbours were uniformly diagnostic and child-related. Alternatively, the contraction may reflect the composition of the sample rather than the concept: if health articles written to a common template (e.g., for search-engine optimisation) account for a growing share of retained ADHD documents, their near-identical contexts would pull measured breadth down even if the meaning of ADHD itself were unchanged. The two possibilities are compatible and cannot be separated here. Either way, the *frustration* comparator's concurrent broadening indicates that the contraction is target-specific rather than a corpus-wide artefact.

The findings also bear on the evidentiary question that motivated the study. Prior evidence for creep in mental-health-related concepts derives largely from psychology abstracts and curated general-language corpora (Vylomova and Haslam, 2021; Baes, Haslam, and Vylomova, 2023; Baes et al., 2023), and Pisl, Bucur, and Podina (2025) have shown that apparent semantic trends in such corpora can be artefacts of shifting discourse composition. The present study addressed both concerns—moving to general web discourse and stratifying by frame—and found no concept creep in ADHD and autism. This does not overturn earlier findings for other concepts, which concerned different terms with different institutional standing, but it reinforces that creep results are corpus- and composition-dependent, and that inferences about cultural change drawn from any single curated corpus should be made cautiously. At the same time, Common Crawl imposes structural constraints of its own (Baack, 2024), a point developed in Section 6.5; the discrepancy between corpora identifies corpus dependence rather than settling which corpus better reflects the underlying culture.

The salience results sharpen this point. Autism's declining prominence in the sampled crawl sits awkwardly beside steeply rising diagnosed prevalence (Zeidan et al., 2022; Australian Bureau of Statistics, 2023) and more than four million TikTok videos hashtagged by June 2026. The most plausible reconciliation is that this discourse growth is concentrated on social platforms and behind robots.txt restrictions, both largely outside Common Crawl's reach (Baack, 2024). Web discourse, in other words, is not a single population: the boom visible on TikTok is not automatically visible—or even present—in web-scale archives.

6.3 From Disorder Talk to Lived Experience

If the meanings of ADHD and autism held broadly steady, the way they are talked about did not. The most reliable change detected in the entire study concerns who and what these terms are used to discuss: ADHD discourse shifted markedly from clinical toward lived-experience framing, and autism discourse moved in the same direction from a substantially higher starting point (a fitted 35.9% lived-experience share in 2014, against 17.5% for ADHD). This asymmetry is theoretically coherent. Identity-first and community framings of autism were established well before the study window, whereas ADHD's popular turn—adult self-recognition, online community formation, and identity-oriented content—is more recent, consistent with qualitative evidence on ADHD online communities (Ginapp et al., 2023), clinical reports of self-diagnosed presentations (Hartnett and Cummings, 2024), the sharp growth in ADHD media coverage (Martin et al., 2025), and accounts of "platformed diagnosis" spilling outward from social media (Alper et al., 2025). On this reading, ADHD is undergoing in the 2020s a reframing that autism discourse partially completed earlier, leaving autism's trend flatter and, after accounting for serial dependence, statistically unresolved.

The compositional shift withstands scrutiny of the instrument that measured it: the temporal-stability audit found lived-experience precision and calibration best in the late period—the opposite of what drifting classifier error would require—and the human-adjudicated passages reproduce the rise independently of the classifier (Appendix Table A.2). The shift likewise vindicates the study's frame-aware design on its own terms. Frame-specific trajectories repeatedly diverged in sign: autism clinical breadth trended nominally upward while lived-experience breadth declined; autism clinical sentiment was positive under the AR(1) model while the lived-experience slope pointed downward. An unstratified analysis would have blended these movements with the changing frame mixture, precisely the conflation that Pisl, Bucur, and Podina (2025) identified in newspaper text. The present results extend that demonstration to web-scale data and to a different kind of composition problem. For *anxiety* and *depression*, stratification must first separate clinical constructs from the ordinary states and non-clinical phenomena (e.g., everyday anxiety, or an economic depression) the same words denote (Xiao et al., 2023); ADHD and autism are diagnostic concepts by definition, so the composition that shifts is the framing of the diagnosis itself—as disorder construct or as identity and lived experience.

The sentiment results add a temporal texture that linear models missed. The flagged ADHD and autism clinical series traced U-shapes with minima between Q3 2018 and Q2 2019, implying declining valence through the mid-2010s followed by recovery through 2026, with the positive valence shift in autism clinical contexts robust to first-order autocorrelation. The turning point falls within the period in which neurodiversity entered mainstream media vocabulary, with US news coverage of neurodiversity and neurodivergent individuals increasing between 2016 and 2022 (Huang-Horowitz, Peterson, and Furey, 2025), and in which first-person ADHD and autism content proliferated on social platforms (Hartnett and Cummings, 2024; Alper et al., 2025). The post-hoc contributors indicate that the recovery is carried by child-, family-, and support-related vocabulary against a persistent background of diagnostic and comorbidity terms. Even so, this is at most gentle destigmatisation, and it is not uniform: the appearance of *dangerous* and *obsession* among the late-period arousal-raising collocates in autism lived-experience contexts shows that alarmed or stigmatising discourse persists.

A further pattern deserves note. From 2018 onward, *adhd* entered the leading negative-valence contributors in autism contexts, mirroring the persistent presence of *autism* in ADHD contexts, and 11,795 corpus documents contained both targets. This is web-general corroboration of the ADHD–autism semantic convergence that Kang, Haslam, and Conway (2025) reported on Reddit from 2019: the two concepts are increasingly discussed together, as co-occurring conditions and as jointly constitutive of neurodivergent identity. The convergence also complicates target-specific measurement, since each concept increasingly forms part of the other's context.

Taken together, these results speak to the therapy-speak debate with which this project opened. The trivialisation concern holds that popularisation erodes the meaning of clinical terms, dispersing them across ever milder and more heterogeneous uses and thereby depriving diagnosed people of precise language for their experiences (Isern-Mas and Almagro, 2025; Spencer and Carel, 2021). For ADHD and autism in general web discourse, the semantic preconditions of that concern are not in evidence: contexts did not become milder, did not broaden, and remained thematically anchored to diagnosis, childhood, and family.

6.4 Measurement Sensitivity and Its Implications for LSC Research

The robustness checks carry an implication that extends beyond this study. Substituting the original SIBling resources for the refined ones changed some substantive conclusions. Under Warriner norms, ADHD intensity declined reliably, a result that, taken alone, would have been reported as vertical concept creep—whereas the NRC–VAD estimate was flat. Under MPNet sentence embeddings, the pronounced XL-LEXEME decline in ADHD breadth attenuated to a null, while autism's flat XL-LEXEME trajectory became a reliable MPNet decline. Two conclusions were common to both operationalisations and can be stated with corresponding confidence: neither target broadened horizontally, and neither showed a reliable linear sentiment trend. The vertical-creep conclusion, by contrast, is measurement-conditional and should be reported as such.

There are principled grounds for preferring the refined measures *a priori*: NRC–VAD offers roughly threefold lexical coverage, multi-word entries, and higher reported reliability (Mohammad, 2025), and XL-LEXEME is a word-in-context model built for LSC detection that represents the target use itself rather than the surrounding sentence topic (Cassotti et al., 2023), a distinction that matters when the analytic object is a term's usage rather than a document's subject matter. But the *a priori* argument does not dissolve the empirical problem: the SIBling dimensions, as currently operationalised, are not measurement-invariant, a possibility its authors anticipated in describing the operationalisations as first implementations (Baes, Haslam, and Vylomova, 2024). This observation may also help explain the mixed findings accumulating in the literature—increases in semantic intensity where creep predicts decreases (Xiao et al., 2023), reversals across corpora (Iacob and Uban, 2026), and composition-dependent trends (Pisl, Bucur, and Podina, 2025)—some portion of which plausibly reflects instrumentation rather than substance. Until the framework's dimensions are validated against human judgements of intensity, breadth, and connotation, LSC studies in this tradition would do well to treat multi-resource robustness checks not as optional supplements but as part of the primary analysis, as practised here.

6.5 Limitations

The most consequential limitations concern the corpus. Common Crawl is neither a complete copy of the web nor a representative sample of it: domain selection follows harmonic centrality, adopted in 2017 (prior crawls relied mainly on externally donated URL seed lists, which had dwindled over time); major social platforms are absent; and rights-holder blocking via `robots.txt` has grown over precisely the years studied (Baack, 2024). Temporal comparability is therefore imperfect, and trends—salience above all—partly reflect changes in what the crawl could see. Findings generalise to quality-filtered, English-language open-web discourse as archived by Common Crawl, not to public discourse at large, and emphatically not to the social platforms where much contemporary ADHD and autism discourse seems to occur.

The frame-aware design inherits classifier error. The lived-experience head performed adequately but weakest (F1 = .760, precision = .704), so lived-experience strata contain clinical admixture, which will have attenuated frame contrasts and blurred frame-specific trends. The temporal-stability audit found no evidence of the more damaging failure mode—non-stationary error manufacturing the compositional trend—but with only 11–23 lived-experience validation positives per period, modest error drift cannot be excluded and the fitted ADHD rise may be somewhat smaller than the apparent 12.8 points. The human gold standard also rests on a single coder: all human labels—pilot, adjudication, and validation—reflect one annotator's judgement, so no inter-annotator agreement can be reported and validation performance is measured against an individual rather than a consensus standard. And although the ACT workflow retained human authority over labels, LLM-assisted annotation may import model-specific biases that human review of high-risk cases only partially corrects. The frame codebook, finally, is specific to ADHD and autism, so the baselines could not be frame-stratified, limiting target–baseline comparisons to unframed series.

Three further limitations should be explicit. The affective measures score lemmatised collocates with context-free lexicon values within the $\pm$5-token windows adopted from the SIBling operationalisation (Baes, Haslam, and Vylomova, 2024), and are therefore blind to negation, irony, and syntactic scope. The comparators are baselines for general negative-affect language, not matched semantic controls. And the design as a whole is observational: it can characterise how discourse changed, but not why.

6.6 Directions for Future Research

Three extensions follow directly. First, cross-platform designs are needed to test the suggestion advanced in Section 6.2 that ADHD and autism discourse has migrated toward social platforms outside Common Crawl's reach: applying the same frame-aware measures to web, Reddit, and short-video corpora over a common window would show whether the semantic stability observed here coexists with creep in platform discourse (cf. de Vries, Batstra, and van Assen, 2025; Aragon-Guevara et al., 2025). Second, the measurement sensitivity identified in Section 6.4 motivates validation studies in which lexicon- and embedding-based intensity, breadth, and sentiment estimates are benchmarked against human judgements of the same contexts, so that resource choice can rest on demonstrated convergent validity rather than plausibility arguments. Third, the substantive account developed here generates testable expectations: if institutional codification protects diagnostic labels from creep, then frame-aware analyses of less codified neighbouring vocabulary (e.g., *neurodivergent*, *masking*, *stimming*, *executive dysfunction*) should reveal greater semantic movement than ADHD and autism themselves; and linking annual discourse measures to diagnostic and prescribing statistics would allow the temporal ordering assumed by the prevalence inflation hypothesis (Foulkes and Andrews, 2023) to be examined directly. Finer-than-annual time resolution, non-English web discourse, and qualitative analysis of the contentious late-period lived-experience contexts flagged in Section 6.3 would each add further resolution to the picture drawn here.

7 | Conclusion

This research project examined lexical semantic change in ADHD and autism across thirteen years of general web discourse, motivated by the concern that the popularisation of mental health language erodes the meaning of the terms on which diagnosed people rely. It constructed a reproducible Common Crawl pipeline spanning 2014–2026, separated clinical from lived-experience framing with a validated hierarchical classifier, and estimated salience, sentiment, intensity, breadth, and thematic-neighbour trajectories using a refined adaptation of the SIBling framework, benchmarked against the framework's original operationalisations.

The hypothesised concept creep did not materialise. ADHD and autism did not drift into milder emotional contexts, and their contexts did not broaden; under the target-aware operationalisation, ADHD contexts contracted markedly, and the thematic core of both discourses—diagnosis, childhood, family—remained stable throughout. What changed was the composition of the discourse: a robust shift from clinical toward lived-experience framing, strongest for ADHD, together with a recent, tentative positive valence shift in clinical contexts and growing entanglement between the two concepts. In sampled open-web discourse, autism became less prominent rather than more, underscoring that the contemporary surge in ADHD and autism talk is unevenly distributed across the digital landscape.

The study makes four contributions. Substantively, it extends concept creep and therapy-speak research to ADHD and autism and provides evidence that two institutionally codified diagnostic concepts resisted the semantic loosening documented for ordinary-language harm concepts. Evidentially, it moves LSC research on mental-health concepts from curated corpora to general web discourse and shows that conclusions do not simply transfer between the two. Analytically, it demonstrates at web scale that discourse composition and semantic change must be separated, since frame-specific trajectories repeatedly diverged from aggregates. Methodologically, its robustness checks show that key SIBling conclusions are conditional on lexicon and encoder choice, identifying measurement validation as a priority for the field.

The broader implication is a reframing of the trivialisation concern. For ADHD and autism on the open web, the words have largely held their shape; it is the conversation around them—who speaks, in what frame, and on which platforms—that has been transformed. Whether the same holds where that conversation is now loudest remains the question this project leaves open, and the tools it provides are designed to answer it.

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Appendices

A.1 Frame Classification Trend Models

Table A.1 reports the descriptive trend models behind the clear-frame lived-experience trajectories in Figure 5.2.

Table A.1: Descriptive annual trend models for clear-frame lived-experience share.

Target	$B(SE)$	β	Adj. R^2	Fitted 2014	Fitted 2026
ADHD	1.07*** (0.15)	0.90	0.80	17.5%	30.3%
Autism	0.59*† (0.20)	0.67	0.39	35.9%	42.9%

Note. The outcome is annual lived-experience share among contexts assigned to one clear substantive frame, defined as lived-only divided by lived-only plus clinical-only. Years are document publication years. $B(SE)$ is the annual OLS slope in percentage points per year; β is the standardised year coefficient. Significance markers: $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$. † indicates a residual-autocorrelation flag; AR(1) sensitivity estimates for flagged rows are reported in Appendix Table A.4. Figure intervals are 95% document-bootstrap intervals. P values are descriptive and uncorrected.

A.2 Classifier Temporal-Stability Audit

Table A.2 reports held-out validation performance for the frozen classifier stratified by period, from the post-hoc temporal-stability audit of the compositional trend. Periods follow the corpus banding (2014–2017, 2018–2022, 2023–2026), which balances the 200-passage validation set at 67/67/66 passages.

Table A.2: Period-stratified held-out validation performance for the hierarchical frame classifier. Brackets give 95% Clopper–Pearson intervals; F1 intervals are stratified bootstrap percentiles.

Period	Support	Precision	Recall	FPR	F1
Substantive target discourse					
2014–2017	45/67	.884 [.75, .96]	.844 [.71, .94]	.227 [.08, .45]	.864 [.78, .93]
2018–2022	46/67	.816 [.68, .91]	.870 [.74, .95]	.429 [.22, .66]	.842 [.76, .91]
2023–2026	50/66	.976 [.87, 1.00]	.800 [.66, .90]	.062 [.00, .30]	.879 [.80, .94]
Clinical framing					
2014–2017	36/45	.897 [.76, .97]	.972 [.85, 1.00]	.444 [.14, .79]	.933 [.87, .99]
2018–2022	37/46	.971 [.85, 1.00]	.892 [.75, .97]	.111 [.00, .48]	.930 [.86, .99]
2023–2026	32/50	.833 [.65, .94]	.781 [.60, .91]	.278 [.10, .53]	.806 [.68, .90]
Lived-experience framing					
2014–2017	11/45	.667 [.35, .90]	.727 [.39, .94]	.118 [.03, .27]	.696 [.43, .88]
2018–2022	12/46	.611 [.36, .83]	.917 [.62, 1.00]	.206 [.09, .38]	.733 [.53, .89]
2023–2026	23/50	.792 [.58, .93]	.826 [.61, .95]	.185 [.06, .38]	.809 [.67, .92]

Note. Support gives positive cases over the per-period evaluation denominator. Substantive target discourse is evaluated on all held-out passages in the period; clinical and lived-experience framing are evaluated among human-substantive passages, following the aggregate validation convention. FPR = false-positive rate.

A.3 Baseline Comparator Trend Models

Table A.3 reports the unframed comparator-term trend models used to contextualise the ADHD and autism target trajectories.

Table A.3: Descriptive annual trend models for baseline comparator trajectories.

Measure	Comparator	$B(SE)$	β	Adj. R^2
Salience	frustration	$-0.0175^{\dagger}$ (0.0167)	-0.30	0.01
Salience	loneliness	0.0056^{*} (0.0021)	0.62	0.33
Salience	sadness	-0.0171^{**} (0.0039)	-0.80	0.61
Sentiment	frustration	$0.0037^{*\dagger}$ (0.0013)	0.67	0.39
Sentiment	loneliness	$0.0044^{**\dagger}$ (0.0013)	0.71	0.45
Sentiment	sadness	-0.0030^{**} (0.0007)	-0.79	0.59
Intensity	frustration	$-0.0004^{\dagger}$ (0.0006)	-0.23	-0.03
Intensity	loneliness	-0.0006 (0.0006)	-0.30	0.01
Intensity	sadness	0.0028^{***} (0.0004)	0.88	0.76
Breadth	frustration	0.0020^{**} (0.0006)	0.72	0.48
Breadth	loneliness	-0.0007 (0.0004)	-0.51	0.19
Breadth	sadness	$-0.0010^{**\dagger}$ (0.0003)	-0.71	0.46

Note. Cells report annual unstandardised OLS slopes as $B(SE)$; β is the standardised year coefficient. Comparator terms are unframed baseline series. Salience uses Common Crawl source year and semantic measures use document publication year. Significance markers: $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$. † indicates a residual-autocorrelation flag; AR(1) sensitivity estimates for flagged rows are reported in Appendix Table A.4. P values are descriptive and uncorrected.

A.4 AR(1) Sensitivity Models

Table A.4 reports AR(1) sensitivity estimates for daggered rows in the trend tables.

Table A.4: AR(1) sensitivity models for autocorrelation-flagged annual series.

Source	Measure	Series	Frame	OLS B	AR(1) B	AR(1) p
Table 5.1	Sentiment	ADHD	Overall	0.0023	0.0037	.147
Table 5.1	Sentiment	ADHD	Clinical	0.0013	0.0036	.218
Table 5.1	Sentiment	Autism	Clinical	0.0022	0.0052	.040
Table 5.1	Breadth	ADHD	Lived experience	−0.0004	−0.0004	.247
Table 5.1	Breadth	Autism	Overall	0.0001	−0.0001	.753
Table 5.1	Breadth	Autism	Clinical	0.0007	0.0002	.755
Table A.1	Frame classification	Autism	Lived vs clinical	0.59	0.23	.370
Table A.3	Salience	frustration	Baseline	−0.0175	0.0160	.538
Table A.3	Sentiment	frustration	Baseline	0.0037	0.0116	.003
Table A.3	Sentiment	loneliness	Baseline	0.0044	0.0070	.009
Table A.3	Intensity	frustration	Baseline	−0.0004	−0.0022	.078
Table A.3	Breadth	sadness	Baseline	−0.0010	−0.0008	.131

Note. Rows are the series marked with † in the compact regression tables. OLS B repeats the primary annual slope; AR(1) B reports the first-order autoregressive sensitivity slope for the same series. The frame-classification row is reported in percentage points per year; all other rows use the original index units per year. P values are descriptive and uncorrected.

A.5 Quadratic Trend Models

Table A.5 reports the quadratic models for annual LSC trajectories whose linear residuals were flagged for autocorrelation.

Table A.5: Supplementary quadratic models for autocorrelation-flagged LSC trajectories.

Measure	Target	Frame	Linear $B(SE)$	Quadratic $B_2(SE)$	Δ Adj. R^2	Vertex
Breadth	ADHD	Lived experience	-0.0004 (0.0006)	-0.0000 (0.0002)	-0.11	outside window
Breadth	Autism	Overall	0.0001 (0.0002)	-0.0002^{**} (0.0000)	0.58	Q2 2020 (inverted U)
Breadth	Autism	Clinical	0.0007^{*} (0.0003)	-0.0003^{***} (0.0001)	0.51	Q2 2021 (inverted U)
Sentiment	ADHD	Overall	0.0023 (0.0014)	0.0009^{*} (0.0003)	0.29	Q3 2018 (U)
Sentiment	ADHD	Clinical	0.0013 (0.0015)	0.0012^{**} (0.0003)	0.60	Q2 2019 (U)
Sentiment	Autism	Clinical	0.0022 (0.0015)	0.0013^{***} (0.0002)	0.65	Q1 2019 (U)

Note. Rows are restricted to annual series whose linear-model residuals were flagged for autocorrelation. Linear B is the annual OLS slope from the primary trend model. Quadratic B_2 is the centred-year-squared coefficient; Δ Adj. R^2 is the adjusted-R^2 gain over the linear model. Vertex labels give the calendar quarter containing the fitted decimal-year vertex when it falls inside the observed annual window; otherwise the vertex is marked as outside the window. Significance markers are descriptive and uncorrected: $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

A.6 Post-hoc Contributor Tables

Tables A.6.1–A.6.3 summarise the main VAD collocate contributors and the breadth diagnostic inspected after the annual LSC analyses.

Table A.6.1: Post-hoc sentiment collocate contributors by target, frame, and period.

Target	Frame	2014-2017	2018-2021	2022-2026
ADHD	Overall	Positive: child, adult, add Negative: disorder, autism, depression	Positive: child, like, add Negative: autism, disorder, depression	Positive: child, like, adult Negative: autism, depression, disorder
ADHD	Clinical	Positive: child, adult, add Negative: disorder, depression, autism	Positive: child, like, add Negative: disorder, depression, autism	Positive: child, like, adult Negative: depression, autism, disorder
ADHD	Lived experience	Positive: child, adult, parent Negative: autism, diagnose, dyslexia	Positive: child, like, adult Negative: autism, anxiety, dyslexia	Positive: child, like, adult Negative: autism, dyslexia, diagnose
Autism	Overall	Positive: child, developmental, family Negative: disorder, disability, syndrome	Positive: child, developmental, family Negative: disorder, disability, adhd	Positive: child, family, developmental Negative: disorder, adhd, disability
Autism	Clinical	Positive: child, developmental, include Negative: disorder, disability, disease	Positive: child, developmental, include Negative: disorder, disability, adhd	Positive: child, developmental, include Negative: disorder, adhd, disability
Autism	Lived experience	Positive: child, son, family Negative: disorder, disability, syndrome	Positive: child, son, family Negative: disorder, disability, diagnose	Positive: child, family, son Negative: dangerous, disorder, adhd

Note. Cells show the three largest preprocessed collocate contributors from each direction.

Table A.6.2: Post-hoc arousal collocate contributors by target, frame, and period.

Target	Frame	2014-2017	2018-2021	2022-2026
ADHD	Overall	Raising: disorder, anxiety, hyperactivity Lowering: add, treatment, patient	Raising: anxiety, disorder, hyperactivity Lowering: add, sleep, student	Raising: anxiety, disorder, challenge Lowering: individual, common, treatment
ADHD	Clinical	Raising: disorder, anxiety, hyperactivity Lowering: treatment, add, patient	Raising: disorder, anxiety, hyperactivity Lowering: add, sleep, narcolepsy	Raising: anxiety, disorder, hyperactivity Lowering: treatment, common, individual
ADHD	Lived experience	Raising: anxiety, suffer, struggle Lowering: parent, student, add	Raising: anxiety, challenge, disorder Lowering: student, add, dyslexia	Raising: challenge, anxiety, struggle Lowering: autistic, dyslexia, individual
Autism	Overall	Raising: disorder, developmental, spectrum Lowering: link, family, individual	Raising: disorder, developmental, spectrum Lowering: people, individual, family	Raising: disorder, developmental, dangerous Lowering: individual, people, family
Autism	Clinical	Raising: disorder, developmental, schizophrenia Lowering: link, center, individual	Raising: disorder, developmental, spectrum Lowering: link, individual, center	Raising: disorder, developmental, schizophrenia Lowering: individual, therapy, link
Autism	Lived experience	Raising: disorder, boy, spectrum Lowering: son, family, people	Raising: disorder, challenge, spectrum Lowering: people, son, individual	Raising: dangerous, obsession, disorder Lowering: individual, people, family

Note. Cells show the three largest preprocessed collocate contributors from each direction.

Table A.6.3: Post-hoc breadth high-distance content words by target, frame, and period.

Target	Frame	2014-2017	2018-2021	2022-2026
ADHD	Overall	disorder, behavior, childhood, impulsivity, inattention	disorder, include, child, symptom, syndrome	disorder, child, autism, learning, learn
ADHD	Clinical	disorder, behavior, childhood, impulsivity, inattention	disorder, include, syndrome, autism, disability	disorder, condition, autism, use, learning
ADHD	Lived experience	disorder, thing, child, parent, school	disorder, autism, student, diagnose, people	disorder, autism, dyslexia, spectrum, autistic
Autism	Overall	lead, need, time, society, channel	world, day, feel, high, thing	lead, research, fact, know, actually
Autism	Clinical	family, individual, research, speaks, government	child, early, new, organization, speaks	speaks, need, developmental, treatment, research
Autism	Lived experience	far, car, actually, attitude, away	world, day, feel, high, work	experience, think, people, program, speaks

Note. Cells show the five most frequent filtered content lemmas among the twenty highest-distance contexts in each cell. This breadth diagnostic summarises high-distance context wording rather than VAD collocates.

A.7 ADHD Neighbour Similarity Evolution

Figure A.1 presents the annual similarity trajectories for the stable ADHD neighbours reported in the Results chapter.

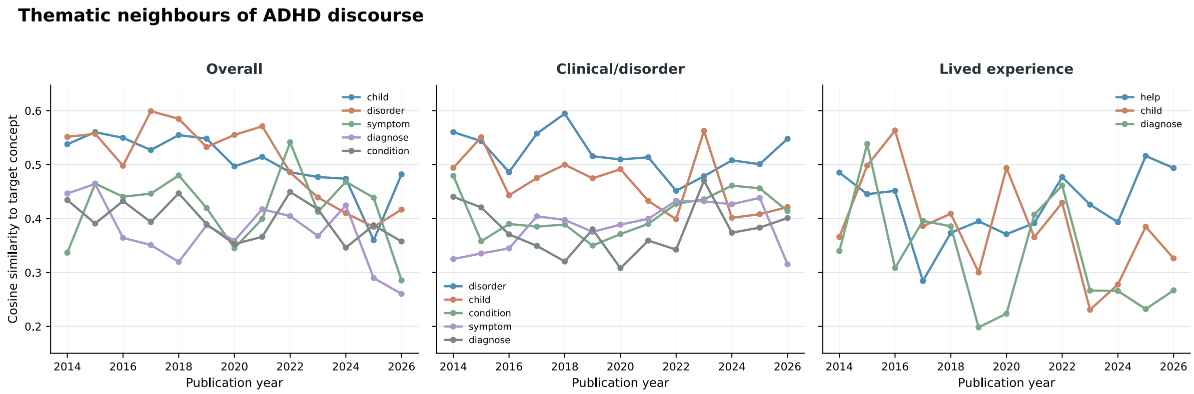

Figure A.1: Annual cosine similarity between the ADHD concept token and stable Word2Vec neighbours in overall, clinical/disorder, and lived-experience contexts. Stable neighbours appeared in an annual top-five list in at least two years and had estimates in at least ten years.